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Review



A survey on authentication protocols of dynamic wireless EV charging

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ABSTRACT

Electric Vehicles (EVs) are considered the predominant method of decreasing fossil fuels as well as greenhouse gas emissions. With the drastic growth of EVs, the future smart grid is expected to extensively incorporate dynamic wireless charging (DWC) systems, a significant advancement over traditional charging methods. DWC, offering the unique ability to charge vehicles in motion, introduces new infrastructures, complex network models and consequently, a massive attack surface. To accomplish the goal of such an enormous smart grid accompanying DWCs, the security of EV charging infrastructures has become a deciding factor. EV charging is vulnerable to cyberattacks as it has many attack vectors and many challenges to combat. Unlike the traditional charging services provided in a typical static charging station, the DWC has a complex network architecture which makes it vulnerable to many forms of cyberattacks. Authentication plays a crucial role in safeguarding the frontline security of this ecosystem. However, within the domain of DWC, the current academic landscape has seen limited attention dedicated to authentication protocols. This background signifies the necessity of a comprehensive survey to cover the authentication protocols of dynamic wireless EV charging environments. This review paper examines the security requirements and the network model of the DWC, providing comprehensive insights into existing authentication protocols by scrutinizing a proper classification. Furthermore, the paper addresses existing challenges in authentication schemes within DWC and explores potential future research tendencies aiming to strengthen the security framework of this emerging technology.

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1. Introduction

The modern world is contemplating EVs as one of the most promising solutions to confront accelerating air pollution. Recent environmental statistics show that the transport sector alone accounts for 16.2% of total global greenhouse gas (GHG) emissions [1]. The largest portion of this GHG emission of the transportation sector is powered by the burning of fossil fuels for road vehicles. As a result, researchers, policymakers, and governments of many countries have embarked on their national strategies for zero-emission transportation. Norway as the world's electric car capital [2], has proclaimed the Norwegian EV policy with their national goal to 100% transition towards zero-emission (electric or hydrogen) vehicles by 2025 [3]. In December 2022, the government of Canada declared to increase the supply of EVs to achieve the goal of 60% of total vehicles to be accommodated by EVs by 2030 [4]. Correspondingly, billions of EVs are predicted to be live on the road in the near future.

To leverage the drastic growth of EVs, electrical vehicle charging infrastructures (EVCI) are also being planted all around the globe. Apart from the technological innovation of EVs, an effective charging infrastructure plays a fundamental role in supporting the wider adoption of EVs [5]. Conductive EV charging or simply charging an EV through an EVCI is a method of charging EVs that involves the physical transfer of electrical energy through conductive connections. This charging method requires a charging cable to be physically connected to the charging station through the charging port [6,7]. According to the global EV statistics, in the year 2021, there were 1.8 million publicly accessible fast chargers in charging points worldwide, as illustrated in Fig. 1. This number shows a growth of 37% in 2021 in comparison to the previous year, despite the pandemic-related slowdown. Public slow charger installation also reveals an average upswing rate of over 60% between 2015 and 2020 [8]. The significant expansion of the charging infrastructure is depicted in Fig. 1. Researchers are actively exploring advanced charging techniques, such as conductive and inductive charging, to cater to the evolving demands of sophisticated lifestyles of users [9] as shown in Fig. 2. These innovative methods aim to provide seamless and convenient charging options for EVs, ensuring that users can easily recharge their vehicles as they go around their daily routines. Unlike conductive charging, inductive charging which is most commonly called wireless charging does not require any physical connection between the EV and the charging equipment. It uses electromagnetic fields to transfer electrical power from a charging utility to the EV without the involvement of any physical cable [10,11]. Energy service providers and EV experts are working together to make this technology match the modern community.

Corporations like WiTricity [12] have been involved in several projects and partnerships to develop wireless charging solutions for EVs and collaborate with Nissan to implant wireless charging technology for the Nissan LEAF electric vehicle [13]. Qualcomm Halo [14], ElectReon [15], and HEVO Power [16] are some industry projects that work on deploying wireless charging systems in various locations, including parking lots and garages, to enable wireless charging for EVs a global service. As illustrated in Fig. 2, EVs can be charged by using a wireless (inductive technique) in various scenarios. These include in motion (dynamic inductive charging) [17,18], at complete rest (static inductive charging) while parked at home or in a public parking lot [19], or during brief idle periods (quasi-dynamic inductive charging), such as at a traffic light or bus stops. The main advantage of the specified technique over traditional conductive charging is the significant increase in the driving range of EVs by enabling continuous charging while on the road [20]. EVs can rely on continuous charging, reducing the need for large and heavy battery packs [21] as dynamic charging reduces the need for large batteries in EVs. DWC systems can be easily integrated with intelligent routing algorithms and enhance energy efficiency, reduce charging station congestion, and improve transportation planning in future smart cities. Moreover, when autonomous vehicles become more widespread, dynamic charging can offer a seamless charging solution that does not require human intervention and leverage this technology to charge while in operation.

In 2009, the Korea Advanced Institute of Science and Technology (KAIST) in South Korea introduced the inaugural commercially operational dynamic wireless charging EV, known as the on-line electric vehicle (OLEV) [22], which garnered substantial media attention [23]. In 2013, the research initiative Inductive Charging was completed successfully, with Flanders' DRIVE (now Flanders Make) leading the project in collaboration with nine companies and two universities, over four years [24,25]. The research project, through the integration of the inductive charging system into various test tracks, conclusively demonstrated the feasibility, safety, and efficiency of wireless charging, both in dynamic and static modes. The results revealed that it is entirely possible to integrate foreign components into conventional road infrastructure, opening up new possibilities for road functionality. After enhancing and optimizing the road structure's design, a real-life project was initiated in Braunschweig, Germany, by Bombardier. Since October 2015, electric buses have been operating in Bruges, Belgium, following a 45-min loop, with a subsequent 12-min inductive charging period.

Dynamic EV charging is a technology that gained significant attention in recent years. Numerous dynamic EV charging projects are currently underway, with a primary focus on heavy vehicles such as buses, trucks, and even passenger cars. Some countries have already made substantial development in this capacity, notably by introducing electrified roads and implementing extensive bus routes that rely on dynamic charging systems. European countries are taking progressive steps towards the electrification of their highway networks, showcasing a commitment to revolutionizing the way of utilizing EVs. This innovative technology has activated a wave of ongoing research experiments aimed at its implementation and further refinement. We highlight some of the key global research projects in the dynamic EV charging domain, reflecting the growing interest and commitment to advancing this transformative aspect of electric mobility. Table 1 serves as concrete

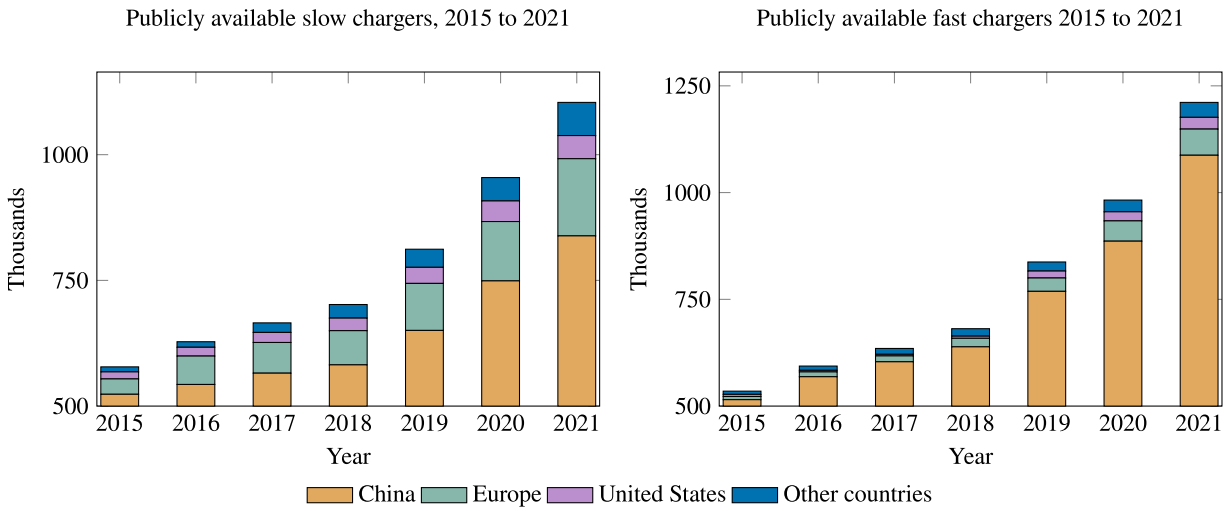


Fig. 1. Global growth of publicly available slow and fast charging infrastructures.

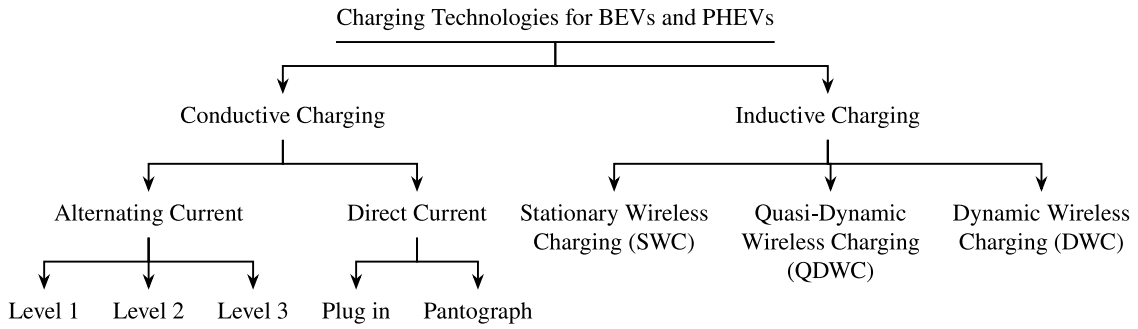


Fig. 2. Classification of charging technologies for BEVs and PHEVs.

proof of the recent progress and practical results of dynamic charging technology adopted worldwide.

Despite advancements in EV technologies, vulnerabilities within the infrastructure render the system prone to cyberattacks. During the dynamic charging sessions, a variety of personally identifiable information (PII) such as EV identity, location information, and payment information is exchanged between the EV and the charging infrastructures [30]. This information can be utilized to derive the location and travel behaviours of the EV. Communication entities involved in DWC charging have the potential for a massive number of threats, including, replay, man-in-the-middle attacks, guessing, eavesdropping and hijacking. Additionally, malicious actors can exploit wireless links between EVs and charging infrastructure to intercept messages transferred from the EV or charging pads by either introducing delays or replaying the intercepted messages to alternative destinations. Furthermore, adversaries can execute impersonation attacks, assuming the likeliness of an EV to engage in dynamic charging without prerequisite payment or posing as a charging station to hijack sensitive information from both the EV and infrastructure, thereby facilitating a network mapping attack. Researchers have proposed many authentication protocols integrating advanced cryptographic primitives, in which the user can authenticate itself to the wireless charging utilities to perform secure communications. However, recent research shows a significant gap in security in EVCI [31–33] and shows that investigations carried out on EV and EVCI security are relatively low. Particularly, concerning the authentication protocols among EVs, chargers, and the broader network in dynamic wireless charging scenarios. Secure authentication is one of the biggest challenges in the DWC system due to its complex architecture. A weak authentication protocol can expose sensitive information about the EV, such as vehicle ID, current location, and charging behaviours, thereby violating the privacy of the EV. Moreover, an attacker who gains access to the system can further propagate their attack towards the power grid, potentially harming the infrastructure. Therefore, implementing robust and privacy-preserving authentication protocols in the DWC system is essential.

1.1. Motivation and our contribution

Ensuring secure and privacy-preserving authentication is one of the most significant challenges in DWC systems. Since dynamic charging is a relatively new technology, there are only a few survey papers available to provide a comprehensive overview of the authentication protocols in the environment. Existing papers such as Jang et al. [34] and Cao et al. [35] cover only the operation and system study on wireless charging systems. They do not review the state-of-art authentication protocols, their strength and limitations in their surveys. Johnson et al. [36] has conducted a review on the cybersecurity aspects of the DWC system covering some vulnerabilities and defences. Raveendra et al. [37] conducted another survey on DWC protocols. However, all these existing surveys do not focus on the types of authentication methods, achieved security measures, and how the cryptographic primitives are utilized to achieve those security measures. Understanding the underlying cryptographic techniques and how they are used to achieve the expected security measures is crucial for the researchers to assess the strengths, and weaknesses of the protocols. Moreover, in the existing research works, there are notable differences in the considered system model, the role of the entities and the theory of operation. As a result, these differences have led the researchers to overlook some crucial security requirements and end up with weak authentication solutions. Therefore, we identified an urgent need to address this

Table 1
Recent studies and pilot projects on dynamic wireless EV charging.

Project, Country (Year)	Collaboration	Remarks
On-Line Electric Vehicle (OLEV) - Korea (2009)	Korea Advanced Institute of Science and Technology (KAIST)	First OLEV bus route was launched in March 2010, next in Sejong (2015) and Gumi (2016). A new route with improved infrastructure began in 2019 in Yuseong District.
Oak Ridge National Laboratory (ORNL) - United States (2012)	Department of Energy's Oak Ridge National Lab, Renewable Energy Lab, Hyundai Motor Group, Virginia Institute of Technology	Has licensed its dynamic wireless EV charging technology, including the polyphase electromagnetic coil and converter, to Brooklyn-based HEVO. Aims to prepare the technology for commercial production [23].
Inductive Charging - Belgium (2013)	Flanders'DRIVE (now Flanders Make), Van Hool, Volvo, Bombardier Transportation, Belgian Road-Research Center	Inductive charging system integrated into traditional roads leading to a practical project in Braunschweig, Germany. Electric buses have been operating in Bruges, Belgium on a loop since October 2015, with a 12-min inductive charging period [26].
Primove Mannheim Germany (2015)	Rhein-Neckar-Verkehr, Bombardier Transportation, Karlsruhe Tech Institute, HESS AG Busses, German Federal Ministry of Transport	A small-scale test using dynamic inductive charging technology, involved two local buses. Tested in the Primove-Mannheim research project to assess the potential cost reductions in public transportation.
Navya Arma Circulator Shuttle Service - United States (2018)	Mcity at the University of Michigan	First automated shuttle service in the US, utilizing two electric shuttles for fixed-route transportation at the University of Michigan. Features a 1-mile loop with two stops, supported by 33-kWh onboard batteries and L2 onboard plugin chargers.
Electreon's Global Wireless-Road - France (2021)	Electreon, VINCI Autoroutes, Gustave Eiffel University	Electreon (Israeli firm), works in France and Norway to deploy its wireless electric roads system for fleet operators. In 2019, a 1.65-km dynamic wireless charging system was completed in Sweden, where a 40-ton electric truck, is powered by the electrified roadway [27].
Wireless Electrified Roadway - United States (2023)	Michigan Department of Transportation (MDOT), Michigan's automotive and energy sectors, Electreon	America's first wireless charging road opens for testing on Nov 2023, utilizes ground-embedded coils for dynamic vehicle charging, promoting EV adoption and aiming for a zero-emission mobility future. The electrified roadway stretches 0.25 miles in Detroit, Michigan [28].
Electric Road System (ERS) - France (2024)	Electreon, Bpifrance, French Ministry of Ecological Transition	Planned for a 2024 launch, this project features 2 km of Electreon's DWC system on the A10 highway. Aimed at decarbonizing fleet transport, It is a blueprint for nationwide ERS deployment, with advancements in charging technology and software, promising significant CO2 reductions for heavy-duty transport [29].

gap. We conducted a survey, motivated by the aim of providing a comprehensive exploration of DWC authentication protocols.

Our survey was initiated by conducting a thorough search on renowned academic databases such as IEEE Explorer, Springer, Elsevier and ACM library, publications spanning the years 2015 to 2024. The priority was to identify state-of-the-art authentication schemes relevant to the DWC context. From an initial pool of 60 research papers, a rigorous filtering process was employed, which considered factors such as recency, alignment with the core research objectives and the significance of the addressed challenges. Methodological validity and the quality of presentation were also crucial criteria. The outcome of this selection process resulted in an appealing collection of nearly 30 papers, including diverse perspectives, methodologies, and contributions. This collection enabled us to produce a comprehensive overview of DWC authentication protocols, including how the authentication techniques evolved, providing valuable insights for both current and future research and development in this field. The preliminary focus of this survey is to present an overview of authentication protocols in DWC systems and how they meet the essential security measures. The major contributions of this paper are as follows.

- Firstly, we present a preliminary understanding of the dynamic wireless charging technology covering the evolution of the technology, system architecture, communication model and theory of operation.
- We discuss the unique challenges of authentication in DWC addressing the anticipated security requirements, the adversarial model, potential security attacks and vulnerabilities, and the countermeasures to achieve robust security. This discussion helps novice readers understand how an authentication solution should align to achieve fundamental security properties within the DWC environment.
- Next, we provide a new classification of DWC authentication protocols by reviewing a range of state-of-art authentication protocols. We present the phases as well as the strengths and limitations of each protocol. Further, we highlight the underlying cryptographic primitives, emphasizing how the security measures are achieved and the involvement of trusted entities in the DWC authentication.
- Finally, we identify the open research problems in the reviewed authentication protocols and elaborate on the potential research directions. This is helpful for researchers in designing authentication protocols to develop contemporary solutions with enhanced security and privacy.

Table 2 illustrates the surveys undertaken on authentication protocols on DWC-related scenarios, and signifies the differences between those studies and our research contribution. As we are covering many crucial aspects which are overlooked in the existing research works, our work might lay a new foundation for novice researchers as well as security professionals and protocol designers in the domain.

1.2. Structure of the paper

The subsequent sections of this paper are structured as follows: Section 1 introduces the fundamentals of dynamic wireless EV charging and a brief idea about authentication mechanisms. Section 2 covers the categories and components of the DWC system, communication model, theory of operation, risks and challenges of authenticating in the DWC environment. Section 3 examines the security fundamentals of the system including an extensive list of security attacks and establishing the security requirements based on the CIA triad. Section 4 provides a comprehensive survey of recent advances in the authentication protocols domain with a new classification approach. This section is the place where existing related works that are carried out in the dynamic

Table 2
Recent studies on authentication of dynamic wireless EV charging.

Description	[35], 2023	[37], 2022	[38], 2022	[36], 2022	[34], 2023	This paper
Outline the foundational elements of dynamic wireless charging systems. (Technology utilization, system components, communication models, and adversary model)	✗	✓	✓	✓	✓	✓
Evaluating the distinct risks and challenges involved with authentication in DWC.	✗	✗	✗	✗	✓	✓
Provide a classification of existing authentication protocols.	✗	✗	✗	✗	✗	✓
Provide a comparative examination of state-of-the-art authentication protocols	✓	✓	✗	✗	✗	✓
Highlight current obstacles within authentication protocols for DWC scenarios and provide insight into prospective future research.	✓	✓	✗	✗	✗	✓
Exploring the different authentication approaches, underlying cryptographic primitives and achieved security measures.	✗	✗	✗	✗	✗	✓

EV authentication domain are discussed in detail with thorough insights about their strengths and limitations. Section 5 explores ongoing research challenges and future directions in this field, while Section 6 concludes the paper by reflecting on its limitations. Table 12 provides a list of abbreviations used throughout the paper.

2. Understanding dynamic wireless EV charging and its authentication mechanisms

Wireless EV charging is still an emerging and relatively new technology for researchers and engineers. This section provides a clear explanation of the system environment and its components of DWC systems. There are two fundamental charging techniques for electric vehicles: conductive charging and inductive charging.

1. **Conductive Charging:** This charging technique allows a direct connection between the EV charger and the EV, with physical contact between the Electric Vehicle Supply Equipment (EVSE) and the battery. Between the several diverse standards, guidelines and rules, there are two categories of charging methods for EVs; power direct current (DC) and alternating current (AC) charging mode. More information related to this method can be found at [39].
2. **Inductive Charging/Wireless Charging:** It is the process of transferring electrical power wirelessly to recharge battery-operated devices without the need for direct physical contact. Utilizing electromagnetic induction, this method enables the charging of portable devices. There are different modes of operation for wireless EV charging. Stationary wireless charging (SWC) is only carried out when the vehicle is idle, for instance, while in parking [40]. DWC is designed with wireless charging pads that are embedded in a small section of the road for example, in the pavement or mounted in the road to facilitate charging for the EV while they are driving [41]. Quasi-dynamic wireless charging (QDWC) occurs while the EV is at transient stops or driving at low speed. In this survey, the main focus is dynamic inductive charging or dynamic wireless charging, in which the charging is facilitated while they are in motion.

2.1. Components of a dynamic wireless EV charging system

DWC is a technology that enables EVs to charge their batteries while in motion, without the need for physical plugs or cables. Commonly referred to as dynamic charging or Wireless Power Transfer (WPT) for EVs. DWC leverages electromagnetic field dynamics to facilitate the seamless transmission of electrical energy from embedded charging infrastructure integrated within road surfaces or other relevant substrates to a dedicated receiving coil or pad mounted beneath the EV [43,44]. The system components of a DWC system are identified as follows:

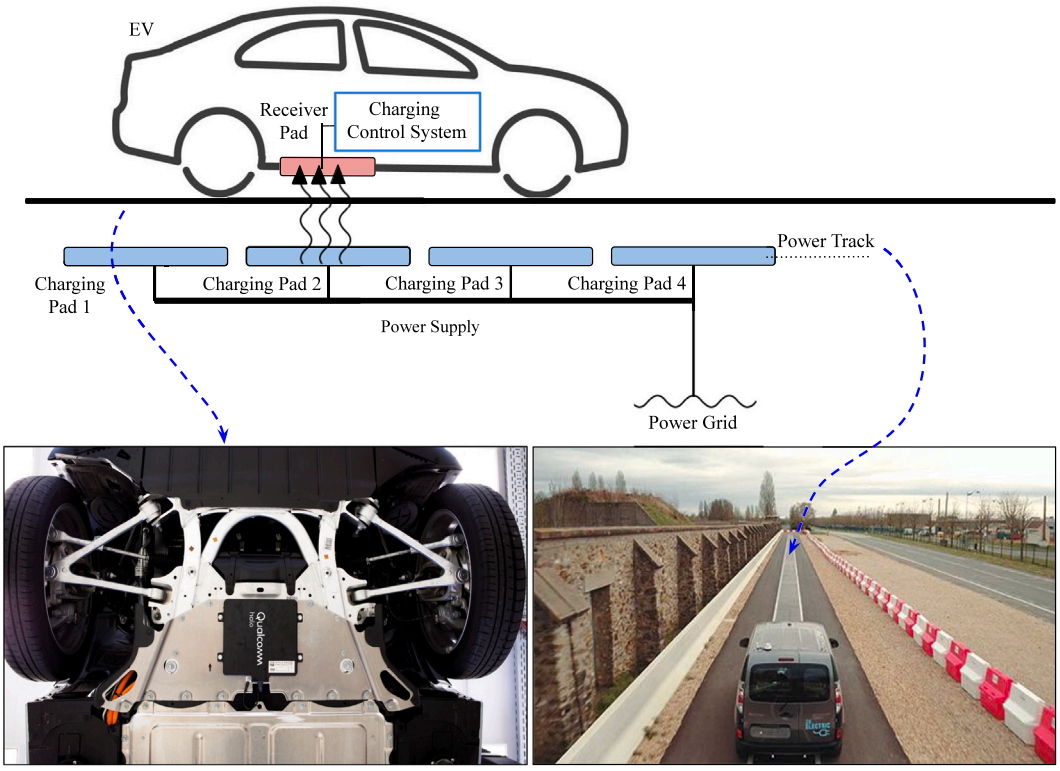
1. Power Supply: It connects a dynamic charging system to the smart grid to receive power.
2. Charging Infrastructure/Power Track: It is embedded into the road surface or the side of the pavement, facilitating the wireless transfer of electric energy when vehicles drive over the track [45,46].
3. Charging Pad (CP)/transmitter (primary) pads: The power track consists of a chain of CPs that use an electromagnetic field to transfer energy wirelessly to the EVs [47].
4. Receiver Pad/secondary pad: It is embedded at the bottom of the EV [48].

The system components mentioned above for DWC are depicted in Fig. 3 given below, enhancing clarity and comprehension. Thus, a dynamic charging system is capable of consistently charging the vehicle while it is in motion through designated charging tracks on a dedicated charging lane [49].

2.2. Communication model of dynamic wireless EV charging

EVs are equipped with on-board units (OBUs) or in-vehicle units (IVUs) for wireless communication facilitating data exchange and control commands. These units serve as the communication interface for EVs, ensuring seamless interaction with the charging components and other infrastructures. Within the network model of a DWC system, apart from the core system components elaborated upon in the preceding subsection, there are additional integral elements, including the roadside units (RSU) and charging service provider (CSP). RSU is a module that also incorporates an OBU to enhance its communication capabilities. It is placed along the road and provides a communication link between the EV [50] and the charging infrastructure and also provides other ancillary services such as real-time traffic and weather information to the EV [51,52]. The CSP is responsible for deploying the infrastructure needed to charge EVs such as charging controllers and charging pads [53].

In the context of wireless communication within the DWC system, Dedicated Short-Range Communication (DSRC) is employed to facilitate the communication between EVs and RSUs. A DSRC is a wireless communication standard designed for short-range vehicle-to-vehicle and vehicle-to-infrastructure communication. It provides a reliable and low-latency channel for exchanging critical data in the vehicular environment between the DSRC-enabled devices [54]. It operates within the 5.9 GHz frequency band, and its engineering is geared towards establishing a reliable and low-latency channel for the exchange of critical data in the dynamic vehicular environment. In addition to the DSRC communication between EVs and RSUs, a backbone network serves as the communication connecting the RSUs to the CP. RSU acts as a communication bridge between EVs and the CP, enhancing the reliability and coordination of data exchange. The EV initiates communication, transmitting messages to the RSU, strategically positioned along the roadway. Equipped with communication capabilities, the RSU



A. Beneath the Electric Vehicle: Receiver Pad Installation

B. Power track embedded to the road

Fig. 3. Components of dynamic wireless EV charging system, for images A and B [42].

receives and relays these messages to the charging pad. Technical data and control commands are transferred through this relay, enabling the charging pad to adjust the power transfer to the receiving coil of the EV based on the requirements. This intermediary role of the RSU ensures efficient and synchronized wireless charging, allowing the EV to communicate its charging needs and facilitate the charging process with precision. This communication architecture is illustrated in Fig. 4, which provides a visual representation of the data flow and interaction between EVs, RSUs, the backbone network, and the CP. This intricate communication framework is crucial for the efficient operation of dynamic charging systems and for establishing dynamic EV charging as a commercial service.

2.3. Theory of operation

Upon entering a power track that facilitates dynamic wireless charging, the EV initiates communication with his corresponding CSP via the cellular network, requesting a DWC session. Then the CSP authenticates the EV and the collaborative process begins with negotiations on the charging rate and agreeing upon pricing for the upcoming charging session [55]. This crucial authentication information, accompanied by the negotiated details, is then relayed from the CSP to the nearest Roadside Unit RSU. Here, the RSU undertakes the authentication of the EV, establishing a secure communication link. The EV provides authentication credentials to the RSU or charging infrastructure, establishing its identity and ensuring authorization to access the dynamic wireless charging service. Simultaneously, the RSU or charging pad authenticates itself to the EV, confirming its legitimacy as an authorized charging point through mutual authentication, establishing a secure communication channel. Following successful authentication, CP starts the charging session and wirelessly releases the amount of energy that was agreed on the charging request with the CSP. EV seamlessly transitions into the dynamic wireless charging session, navigating along the power track while maintaining continuous authentication mechanisms. Smooth handovers between various charging pads, real-time adjustments to power delivery, and the subsequent processes of session termination and payment authorization remain pivotal elements. After the completion of the charging session and payment authorization, the EV exits the charging zone, resuming standard driving operations.

The EV initiates communication through the cellular network, reaching out to its designated Charging Service Provider (CSP) to request a dynamic wireless charging session. Following this, the CSP authenticates the EV, and the charging rate and pricing are determined for the upcoming session. The authentication information, along with other parameters, is then relayed from the CSP to the nearest RSU. The EV provides authentication credentials to the RSU or charging infrastructure to establish its identity and start the charging session; upon successful authentication, the RSU signals the nearest CP to begin the session. Additionally, the EV provides authentication credentials to the CP to prove the identity and once authenticated, the CP starts the charging session and wirelessly releases the pre-agreed requested amount of energy. As the EV moves along the power track, continuous authentication mechanisms ensure a secure connection with the charging infrastructure. A typical power track includes a series of charging pads and a set of RSUs, allowing the EV to seamlessly transition from one CP to another and from one RSU to the next. This continuous authentication and communication throughout the handover process ensure uninterrupted charging as the EV progresses along the track. Moreover, the charging infrastructure continuously monitors the state of the EV and adjusts the power delivery in real time to match the needs of the EV. Once the EV has received the full amount of energy it requested, communications with the charging infrastructure signal the end of the session, with authentication credentials possibly checked again to secure the session's termination. Following this, the charging

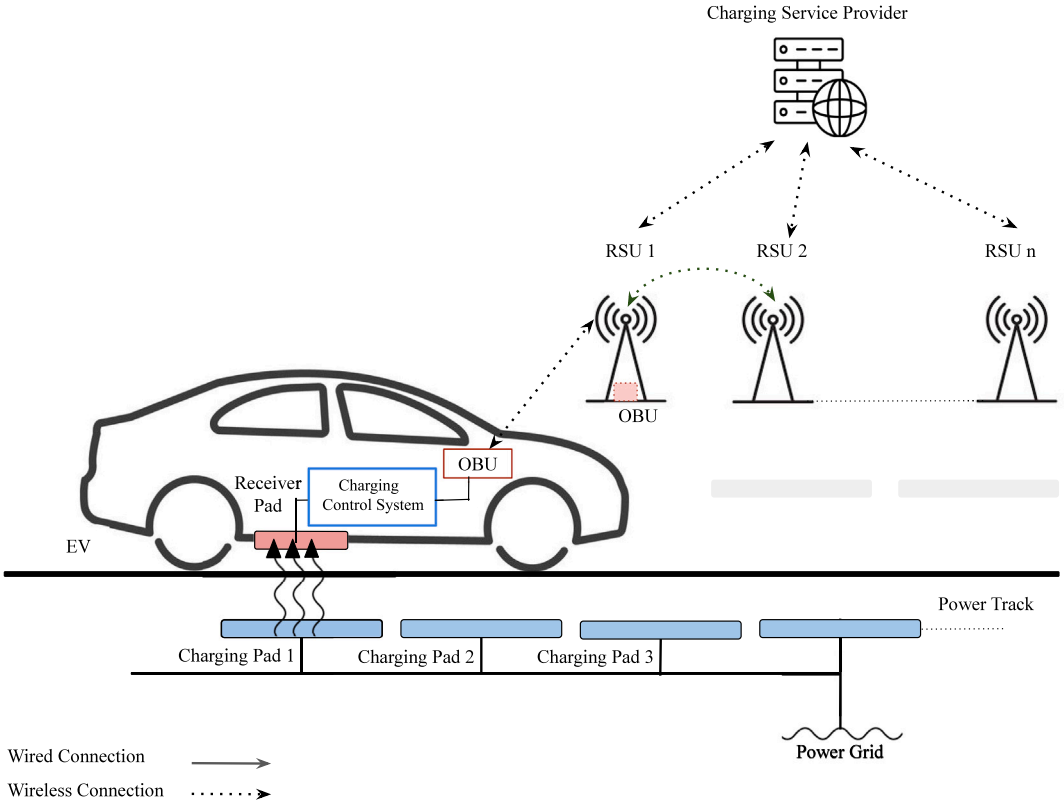


Fig. 4. Communication model of a dynamic wireless EV charging system.

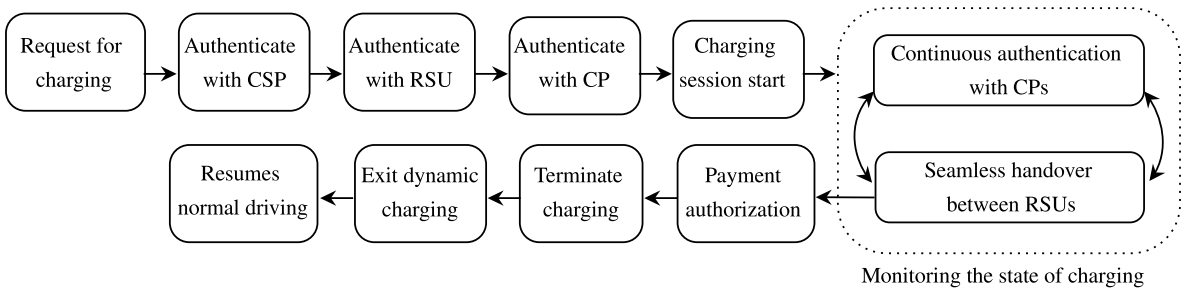


Fig. 5. Dynamic wireless charging - Theory of operation.

infrastructure communicates with the CSP to authorize payment for the charging service provided. Finally, with the charging session completed and payment authorized, the EV exits the charging infrastructure and resumes normal driving operations. Fig. 5 illustrates the theory of operation in a wireless session, systematically representing the sequence of steps involved from the initiation of a charging request to the termination of the session.

2.4. Exploring the complexities of authentication in dynamic wireless EV charging

The National Institute of Standards and Technology (NIST) defines authentication as the process of verifying the identity of an entity, such as a person, device, or system, to ensure that it is indeed who or what it claims to be [56]. Conventional authentication scheme involves presenting credentials, such as passwords, tokens, or biometric data, and validating these credentials to grant or deny access to a specific system, service, or resource. Authentication, in the context of dynamic EV charging, refers to the process of verifying the identity of various entities involved in the charging system, including the EV, the CP, the RSU, and potentially the CSP. It is a crucial aspect of ensuring secure and reliable communication and interaction between various components within the charging infrastructure. It ensures that only authorized and legitimate entities can access and participate in the dynamic charging system, thereby preventing unauthorized access and enhancing security. In contrast to conventional authentication schemes, the process of authentication in a DWC system presents unique challenges and significant risks due to the distinctive characteristics of its operational environment and deployment.

EVs are constantly in motion, necessitating seamless communication and secure authentication as they maintain connections with charging pads embedded in the road. The seamless handover of EVs between RSUs is a critical aspect to ensure uninterrupted power transfer while maintaining a secure connection during the transition from one RSU to another. At the same time, real-time authentication is essential to accommodate the high-speed movement of the EV during dynamic charging. Authentication protocols need to be extremely fast and efficient, allowing the EV to seamlessly connect to new charging pads as it moves. Achieving low-latency authentication can be technically challenging, especially when considering cryptographic operations, secure

Table 3
Security requirements for dynamic wireless charging systems.

Security requirement	Description
Confidentiality	Ensures the privacy of information related to EVs and charging stations, preventing unauthorized access to sensitive data.
Authentication	Verifies the legitimacy of participants and data sources in the DWC system to prevent unauthorized access.
Integrity	The data transmitted throughout the service must remain unchanged and accurate, safeguarding against unauthorized modifications.
Non-repudiation	Enables accurate identification and prevents denial of transmitted messages, ensuring accountability.
Availability	Maintenance of network functionality and consistent service delivery, even under adverse conditions or high demand.
Location privacy	Protects the location information of EVs to prevent unauthorized tracking. The system should ensure that the location data of EVs using the charging infrastructure is not accessible or traceable by unauthorized entities.
Anonymity (Optional)	Conceal the identity of EV users to protect their privacy during transactions and interactions within the DWC system.

key exchange, and other security measures. Many components in dynamic charging systems, such as sensors and communication modules, are resource-constrained. Achieving a delicate balance between robust security measures and the limitations of resource-constrained devices is a key technical challenge [57].

Designing authentication mechanisms that are secure and suitable for resource-constrained devices, including lightweight encryption methods and efficient key management, is crucial. Wireless communication is integral to dynamic charging, but it introduces security risks such as eavesdropping, jamming, and replay attacks. Ensuring secure wireless communication is vital. Advanced technical challenges include implementing strong encryption, secure key exchange, and intrusion detection to protect against wireless security threats. Privacy preservation of an EV's real-time location data is another crucial factor as it is highly sensitive information. An advanced challenge involves anonymizing location data while maintaining the efficiency of authentication and charging processes while maintaining the balance between privacy and functionality. This balance is essential to safeguard EV owners' identities, protect against theft, and ensure secure and efficient charging. Considering the aforementioned risks and challenges associated with DWC systems, it is essential to implement a sophisticated set of authentication protocols that surpass the capabilities of existing traditional authentication schemes to make it fit for this novel echo system. These protocols should be carefully crafted to ensure the efficient, privacy-preserving and secure authentication of system entities. Addressing these advanced technical challenges is crucial for the successful implementation and adoption of dynamic EV charging technology in smart transportation systems.

3. Security fundamentals in DWC systems

3.1. Security requirements

Identifying security requirements for DWC systems is essential to safeguard against potential cyberattacks and data breaches. These requirements include ensuring the confidentiality of sensitive data exchanged between vehicles and infrastructures, authenticating all system participants to prevent unauthorized access, maintaining the integrity of the transmitted data, ensuring non-repudiation for accountability, and guaranteeing system availability even under adverse conditions. Furthermore, location privacy and user anonymity in these systems, are essential for preventing unauthorized tracing and identification of users. The basic security requirements that should be enforced to achieve system security and privacy are illustrated in Table 3

3.2. Security threats and attacks in DWC systems

Given the complex nature of DWC systems, they are vulnerable to a wide range of security threats. These attacks, primarily against critical aspects such as privacy, payment schemes, and authentication and key management, are systematically catalogued in Table 4. The implications of these threats are significant as they are compromising the deciding factors of continuity and reliability of DWC operations. Such attacks could directly affect the continuity and reliability of the DWC process, posing significant service disruptions. Vulnerabilities in DWC infrastructures like hardware damage may lead to significant risks including the potential loss of sensitive data and the disruption of continuous service availability. Therefore understanding and mitigating these vulnerabilities is essential to ensure the secure and efficient operation of DWC systems.

3.3. Adversarial model of DWC

In the DWC system, the main components—EVs, CPs, RSUs, and CSPs are considered untrustworthy and curious entities that may exhibit malicious behaviour. The primary objective of adversaries is to gain unauthorized access to the DWC system, aiming to achieve the following:

1. Inference of communication: Adversaries seek to infer the communication between EVs and the CSP, specifically targeting charging process parameters.
2. Packet interception and replay: Adversaries may intercept and replay packets to trigger actions by EVs or CSP.
3. Unauthorized access gains: Successful attacks may result in free charging rides or the identification of vehicles, which could compromise customer privacy.

Adversaries within the DWC system possess several capabilities, including:

1. Message Composition: Adversaries can create new messages within the network.
2. Replay Attacks: They can intercept and replay messages.
3. Message Interception: Adversaries can intercept messages exchanged between EVs and the CSPA.
4. Message Forgery: The capability to forge messages is also within the range of the adversary.
5. Network Monitoring: Adversaries can connect to the DWC network and monitor traffic.

The threat model assumes that the adversaries lack the cryptographic keys required to decrypt securely encrypted messages, but their focus may be on exploiting vulnerabilities within the authentication

Table 4
Possible security attacks on DWC Systems classified according to the CIA triad.

Attack type	Description	Property			Vulnerability	Potential countermeasures
		C	I	A		
Eavesdropping	Illegally obtaining confidential data from network communications	●			Poor data transmission security	Encryption, VPNs, Secure Protocols
Guessing attacks	Deducing user identities from intercepted messages	●			Weak authentication protocols	Strong authentication, Strong password policies
Free rider	Listening and using messages to charge without paying or authentication		●		Weak message security	Secure Message Authentication, Nonce
Double-spending	Spending the same token more than once during authentication		●		Flawed transaction logic	Unique transaction IDs, Blockchain Technology
Impersonation	Vehicle or server impersonation using false identities	●	●		Identity verification flaws	Strong authentication, Digital certificates
Replay attack	Iterating through past transactions for unauthorized access	●	●		Inadequate session management	Timestamps, Nonce, Session tokens
MITM Attack	Secretly relaying and possibly altering the communication between two parties	●	●		Communication interception	Secure communication channels: TLS, Strong authentication
Privileged-Insider	Trusted insiders exploiting secret credentials	●	●		Insufficient access controls	Role-based access control,
Known key	Linking authenticated keys to extract information	●	●		Key management vulnerabilities	Regular key rotation, Strong cryptography
Session Linking	Connecting sessions to reveal credentials with simple algorithms	●	●		Predictable session tokens	Randomized session IDs, Session management
Message tampering	Modifying message structure to disrupt system operations		●		Lack of message integrity checks	Hash functions, Digital signatures
Injection attack	Sending legitimate messages to gain control and transmit malicious content		●		Unvalidated input handling	Input validation, Parameterized queries
Denial-of-service	Preventing legitimate user access to server resources			●	Resource management weaknesses	Rate limiting, Redundant components to handle excess traffic.
Jamming	Disrupting wireless connections between EVs and network entities			●	Issues with wireless communication interference	Frequency hopping to evade jamming, Threshold-based jamming
Sybil attack	Creating multiple fake identities to subvert the network		●		Identity verification flaws	Strong authentication, Identity verification
Masquerading	Pretending to be a different entity to gain unauthorized access	●	●		Authentication protocol weaknesses	Secure authentication protocols

Legend for table: C-Confidentiality, I-Integrity, A-Availability.

scheme and cryptographic mechanisms. Adversaries may attempt to exploit these vulnerabilities as part of their efforts to compromise the DWC system's security. Poorly designed authentication protocols may provide an opportunity for adversaries to gather information for further attacks, including tracing and replay attacks. For instance, they could use this information to trace the movements of EVs or conduct replay attacks, potentially leading to unauthorized access or data compromise. The threat model in this context is articulated based on the Dolev-Yao model [58], which provides a foundation for understanding and addressing potential vulnerabilities and attacks within systems.

4. Recent advances in authentication protocols of dynamic EV charging

Throughout the research, it became evident that authentication protocols designed for these DWC systems must address the unique challenges and risks inherited by the context it is operating. In response, a new classification of authentication protocols in the DWC domain is also composed which would help future researchers to contemplate when designing authentication protocols. This classification depicted in Fig. 6, to the best of our knowledge, represents the first systematic attempt at categorizing the authentication protocols in dynamic wireless EV charging. This section is dedicated to presenting the new classification of authentication techniques. Each subsection provides a comprehensive overview of research conducted by various authors between 2015 and 2023 in this domain and highlights the limitations. The comprehensive breakdown of each layer and its subdivisions commences from this point, providing a thorough explanation.

4.1. Authentication protocols for charging in DWC

During the evaluation of authentication protocols designed for charging in the context, they are classified into four distinct subcategories, each designed to address specific core properties. These subcategories encompass the following:

1. Privacy-preserving authentication protocols
2. Anonymous authentication protocols
3. Lightweight authentication protocols
4. Efficient authentication protocols

Every subsequent subsection is dedicated to interpreting the realization of requirements and offers a comprehensive exposition of the recent research works within each of the aforementioned subcategories.

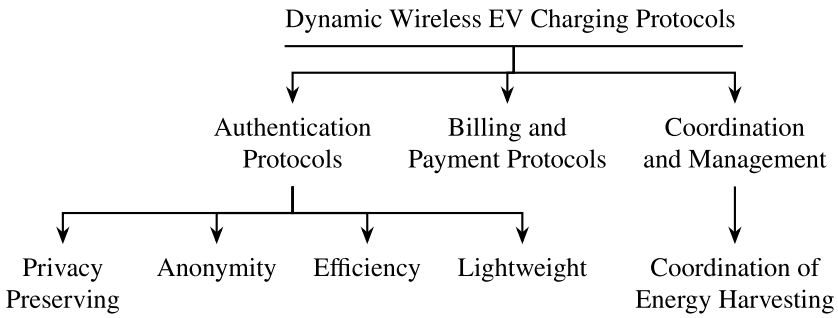


Fig. 6. Classification of authentication protocols in dynamic wireless EV charging.

4.1.1. Privacy preserving authentication protocols

A privacy-preserving authentication protocol in a DWC system is aimed at balancing the authentication needs of EVs with the necessity to protect the privacy of EV users. In DWC systems, where the process of charging EVs often requires scheduling and frequent interactions between EVs and charging infrastructure, the challenge is to verify the identity of the EV without disclosing sensitive information that could compromise user privacy. This protocol allows for secure authentication of the EV to the charging providers while ensuring that the personal details, locations, and charging patterns of the EV users remain confidential. It safeguards against privacy concerns associated with traditional authentication methods and aligns with the principles of connected and electrified transportation systems.

Hussain et al. [59] formulated a privacy-preserving authentication framework based on the assumption that the power track is divided into multiple segments, and the billing process operates on a segment-by-segment basis. In this setup, it is assumed that each segment of the power track provides a consistent amount of energy. A key disadvantage of segment-by-segment billing is that it enables free riders to charge before and after authenticated EVs by utilizing the associated handover delay from one segment to another. Additionally, the examination of security issues is only briefly detailed.

Li et al. [60] proposed Portunes+, an authentication scheme that uses pseudonyms to achieve location privacy, by using pseudonyms and symmetric keys together with spatiotemporal locations of EV. It consists of three main phases key pre-distribution, authentication and accounting. To get rid of the computational overhead of calculation of key sets and secure dissemination of them, authors have used the key pre-distribution phase to be performed before the charging session, for instance: while parked at home. In the authentication step, each EV is assigned appropriate keys and a pseudonym before entering the charging section and is authenticated with charging pads without revealing the true identity of the EV. Finally, in the accounting phase, the PO collects all information from its CPs and forwards it to the CSP for billing. However, in this scheme, CSP is considered a trusted entity to know the location of the EV and each CP is storing the pseudonyms belonging to the EV. Crucially, CP is storing the pseudonyms of the EVs which may be used with the other CPs in subsequent charging segments, which makes the system vulnerable. CP is aware of the linking map between the pseudonym and the real identity which may lead to a privacy violation. The authors conducted a performance evaluation in a test-bed setup with a Raspberry Pi2 Model B using the standard library Crypto++ 5.6.2.

Pazos-Revilla et al. [61] proposed a physical layer-assisted scheme utilizing auto-correlation demodulation and hypothesis testing for DWC. This scheme aims to ensure secure identity authentication, and payment, and safeguard the location privacy of users. The suggested scheme can achieve complete anonymity, preventing any individual from obtaining information regarding the locations of drivers. The scheme operates at the physical layer and uses the shared secrets between the EVs and CPs. The findings demonstrate that the scheme successfully maintains location privacy while remaining efficient and secure against various types of attacks, such as jamming and eavesdropping [38]. To evaluate the computation overhead of the system, the Crypto++ 5.6.2 library has been used.

Roman et al. [62] in 2020, proposed an authentication and authorization protocol for vehicular ad-hoc networks (VANETs) that operate in the cloud and fog environments that utilize short signatures, one-way hash functions, blind signatures based on bilinear pairing to achieve anonymity, integrity and privacy. The protocol achieves confidentiality and accuracy of messages, enables mutual authentication between system components, and ensures the anonymity of EVs as it is dependent on downloadable tickets which are purchased offline from a charging service company before the charging session and signed blindly by the system. This scheme is capable of monitoring and controlling a large number of charging stations effectively. Formal security verification is performed using the AVISPA tool. The protocol is resistant to several types of attacks (MITM, Impersonation, Double Spending, Masquerade attack, etc.), but it is still vulnerable to a DoS attack and lacks adequate mutual authentication techniques. As authors are using bilinear pairing a nontrivial computational overhead will be appended and a complete negotiation process requires 11 times of information exchange which is disadvantageous in fast-paced environments like DWC. In addition, necessary procedures for exchanging primary data between communication entities have not been considered in the protocol.

Roman et al. [63] in 2021, proposed an authentication and authorization protocol designed specifically for fog and cloud-based CWD-WPT charging stations. The protocol incorporates the use of hash functions and a chaotic map-based encryption protocol for key generation, validation, and distribution in a cloud-based CWD-WPT loading station. This is considered a state-of-the-art research work that used chaotic maps to ensure the security properties of the aforementioned systems. Protocol utilizes a Chebyshev map-based cryptosystem to achieve privacy, integrity and anonymity. Additionally, it incorporates a payment mechanism that employs blind signature-based tickets for secure transaction processing. As the tickets are purchased offline, the scheme preserves anonymity. However, this protocol encountered issues with a denial-of-service attack and inadequate mutual authentication mechanisms.

Abouyoussef et al. [64] proposed a state-of-art privacy-preserving networking strategy using blockchain technologies to achieve two main challenges, fast authentication and two-way communications between the CSP and EV. The framework is designed to perform privacy-preserving authentication, manage charging operations, and handle billing processes. In the general structure, a group signature scheme has been implemented on top of a blockchain to achieve the anonymity of the participating entities. A combination of back-channel and blockchain transactions among the entities has been used to reduce the computational overheads. The authors have also presented a novel random number generation technique that provides the necessary interaction between the functionalities and the entities (EV and the CPs). This novel random number generator approach is completely decentralized and ensures that each EV can generate distinct random numbers throughout its usage period. This strategy enables two-way

Table 5
Privacy-preserving authentication protocols in DWC.

Ref, Year	Used Primitives	Achieved properties	Entities	Phases	Limitations
[59] 2015	Hash chain Pseudonymous XOR	+ Conditional privacy + Lightweight mutual authentication	EV-OBU CSP CP VANET authority	1. System initialization 2. TRM installation 3. Pseudonyms assignment 4. Direct mutual authentication	- Assume each segment of the track provides an equal amount of energy. - Security analysis lacking details. - Free-riding is possible.
[60] 2017	Pseudonyms MAC Symmetric key cryptography	+ Location privacy + Allows EVs to charge in different sections and get a single bill + Fast authentication		1. Key pre-distribution 2. Authentication * CSP and * CPs 3. Accounting	- Considerable gap between CPs expected which degrades the performance.
[61] 2018	XOR Hash functions Auto correlation demodulation and hypothesis testing	+ Location privacy + Physical-layer based authorization + Fast authentication and high authorization rate + Full anonymity	EV CSP CP RSU Bank	1. Purchasing of coins 2. Charging request 3. Keys distribution and hierarchical authentication * CSP-RSUs * CSP-EVs * RSU-CPs-EVs 5. Authorization	- Focus only on charging management, no consideration on data protection techniques.
[62] 2020	ECDH Blind signatures Bilinear pairing Hash chain	+ Privacy, integrity and availability + Anonymity	EV, CP CSP, RSU Fog server	1. Initialization 2. EV registration 3. Tickets purchasing 4. Charging request	- Vulnerable to DoS attacks. - Lacking adequate mutual authentication techniques. - Lacks secure primary data exchange.
[63] 2021	Chaotic map (Chebyshev) Blind signature	+ Privacy, integrity + Anonymity		1. Initialization 2. EV registration 3. Ticket purchase 4. Charging request	- High computation cost. - Less suitable for resource constraint environments.
[64] 2021	Blockchain Random number generation mechanism	+ Charging coordination + Authentication and billing	EV CSP CA CP Bank	1. Initialization 2. Charging request 3. Scheduling 4. Authentication 5. Charging, payment	- Scalability, performance of blockchain technology in the context of DWC has not yet been properly tested and evaluated.
[66] 2023	ECDH Blind Signatures Bilinear pairing Hash Chain	+ Trust managed authentication via a trust model + Anonymity via offline tickets	EV, CP RSU, TA Fog server Cloud-CSS	1. Initialization 2. EV registration 3. Tickets purchase 4. Charging request and authentication	- Heavy cryptographic leads to higher processing times - Blockchain and trust model adds latency in transaction processing. - TA and CCS are considered trusted parties.

communications between entities while achieving anonymity and data unlinkability. The scheme was tested for efficiency concerning block generation time, authentication time, and data storage overhead and security was tested upon the user anonymity, data unlinkability, and data tamper-proofness. However, the scalability and performance of blockchain technology in the context of DWC systems have not yet been properly tested and evaluated [65].

Roman et al. [66] formulated a state-of-the-art authentication and access control protocol for DWC systems employing a trust model. As the cryptographic primitives bilinear pairing and trust management, Elliptic-curve Diffie–Hellman (ECDH), blind signatures and hash chains are used as the main building blocks of the protocol. In addition, the protocol comprises an offline ticket-purchasing mechanism to achieve anonymity. In this protocol, TA and CCS are considered trusted parties and used to store EV identification and bank details for ticket purchases. This setup concentrates sensitive information within specific entities, increasing the risk of a single point of failure that, if compromised, could lead to significant privacy breaches and financial fraud. Integrating a blockchain and trust model introduces additional latency in transaction processing, as each transaction must be verified and recorded across multiple nodes in the network. In a resource-constrained and fast-paced environment like DWC, these factors signify the challenges the protocol could face in maintaining efficient operation and real-time transactions. A summary of the privacy-preserving authentication protocols of dynamic EV charging is given in Table 5.

4.1.2. Anonymous authentication protocols

The communication that takes place between the EV, RSU and the charging utilities during a DWC session includes, a variety of personal information such as unique user/EV identity, state of charge, location information, and payment information. This information can be utilized to deduce the location and travel behaviours of the EV. Communication entities involved in dynamic EV charging, have the potential to massive number of threats, including impersonation, replay, man-in-the-middle attacks, guessing, and hijacking. Furthermore, the CSP is normally considered a trusted, but curious entity, which may try to expose the EV user's charging patterns or yielded behaviours to third parties for financial gain. As a strategy, anonymous authentication enables EVs to verify their identity with CSP, charging pads and RSUs without disclosing their true identity. By concealing their true identities, users can be assured that their sensitive information and their charging behaviours remain safeguarded in the DWC ecosystem.

Hamouid et al. [67] proposed a protocol that mainly focuses on achieving anonymity while maintaining fast and lightweight EV authentication by leveraging verifiable encryption, authenticated pairwise keys, and coin hash chains. For the authentication between EV, CSP and charging infrastructure it uses a verifiable pseudonym instead of using the real identities of the user which ensures the privacy of the users. Authentication between EVs and charging pads is performed without the involvement of the CSP and as it requires only one message exchange between the entities which simply perform the verification using a hash chain makes the scheme lightweight and fast. This scheme incorporates conditional anonymity, enabling the traceability of an EV only by the

trusted authority for revocation in the identification of a malicious EV. However, the protocol completely depends on a trusted authority for registration and generates pseudonyms for each EV in every charging session. EV should process this pseudonym to communicate with CSP for charging. The continuous pseudonym requests from various EVs throughout the day can overwhelm the trusted authority, exhausting its resources and efficiency. Besides that, this protocol lacks any security analysis evaluation to verify the security of the protocol.

Kilari et al. [68] proposed a permit-based mechanism (PBM) strategy for authentication that is capable of relocating misbehaving EVs. To achieve anonymity, blind signatures are used along with other cryptographic techniques. For the authentication, the EV should process an anonymous permit issued by a trusted third party (TTP). To obtain a new permit, it is required to have a valid receipt issued by a CSP after a successful charging session. The revocation occurs based on a complaint from a charging entity. The authors argue about the security and privacy of the solution. Specifically, the scheme ensures that a CSP cannot trace back any charging session of the EV, and third parties are unable to independently compromise the anonymity of the EV. The scheme is resistant to application-level denial of service attacks.

Xu et al. [69] formulated a novel EV charging system empowered by blockchain technology, to prevent using third-party platform interventions. In cases where authentication schemes within DWC systems prioritize anonymity, the presence of a robust revocation technique becomes even more crucial. This mechanism effectively handles misbehaving users, thereby minimizing the ability for malicious insider attacks, enhancing security and establishing proper security. The system ensures the establishment of a multiparty security system, between the EVs and EVSPs. Furthermore, zero-knowledge proof and ring-signature superposition are utilized to achieve complete anonymity, undeniability and unforgeability. A summary of the anonymous authentication protocols of dynamic EV charging is given in Table 6

In 2023, Abouyoussef et al. [70] presented a protocol utilizing blockchain technology and an enhanced K-times group signature for ensuring user anonymity which prevents data unlinkability, and protects against DoS attacks. Through a case study in Nashville, TN, USA, it is demonstrated that this strategy can accommodate all publicly charging EVs in the city within a 30-min DWC coordination window, all while providing the necessary security and privacy protections. However, the approach comprises some trade-offs, including high computational and communication demands, which may not be deployed in limited resource environments.

4.1.3. Lightweight authentication protocols

A lightweight authentication protocol in a DWC environment refers to an authentication scheme designed to minimize computational and communication overhead while providing the necessary level of security. It primarily focuses on minimizing computational and communication overhead, making it well-suited for resource-constrained devices like OBUs and EVs. These protocols are designed to ensure that authentication processes are not only fast but also optimized for minimal resource usage. These protocols aim to provide real-time and rapid authentication, critical for the dynamic nature of EV charging processes, without causing significant delays. While lightweight, they must maintain robust security to safeguard against various threats, ensuring trust and reliability in the charging system. Additionally, these protocols prioritize user privacy by limiting the disclosure of sensitive information during authentication.

Raveendra et al. [71] in 2021 proposed an elliptic curve and hash chain-based authentication scheme for DWC authentication which can mitigate a range of security vulnerabilities. However, this scheme is capable of achieving only partial privacy preservation for the EV. The study lacks mechanisms to reduce the amount of communication needed during the sign-in process. In addition, the eavesdroppers can still intercept the communication between the EVs and RSUs at the physical layer when authentication messages are converted into bits, which introduces many security threats. Protocol also suffers from fog-server impersonation and replay attacks.

In 2022, Raveendra et al. [72] presented a lightweight, privacy-preserving authentication scheme that comprises only hash functions, XOR operations and concatenations as basic cryptographic primitives to ensure their main goal is to achieve lightweightness. The scheme proposed novel features to the domain like seamless EV handover and dynamic addition of RSUs and EVs. Using random oracle model and scyther tool authors have performed both formal and informal security analysis. This scheme is capable of providing a certain level of anonymity, location privacy, seamless handover of EVs in between RSUs and security against the following major attacks; impersonation, privileged insider, MITM and DoS (for a certain level). However, the CSP is considered a trusted authority which might be unrealistic in real-world scenarios and may lead to the violation of the privacy of data and the anonymity of the users. The suggested revocation protocol performs only a self-revocation for the EVs, which behaves as a normal service un-subscription. Protocol lacks any mechanism where the authority or the CSP to revoke the malicious EVs from the system, to prevent further attacks. As the scheme requires multiple rounds of mutual authentication, there is a practical question of the suitability of the scheme for fast-phase environments like dynamic EV charging.

Alshaeri et al. [73] proposed a lightweight authentication and authorization protocol constructed on physically unclonable functions (PUFs). Linear feedback shift registers (LFSRs)-based stream cipher (Espresso) is used to protect data integrity and information hiding throughout the communication between EV and CP. The proposed scheme has been evaluated based on communication and computation overhead metrics, demonstrating the use of lightweight primitives for fast authentication and privacy preservation. To achieve efficient and real-time charging, authors have implemented espresso stream cipher together with blockchain technology. However, the use of authentication tokens adds a considerable computational cost to the system. Security validation of the protocol is performed using the AVISPA tool. The authentication process between the EV and CP takes 0.051 ms. However, the authors have tested their protocol in a 2.8 GHz Dual-Core Intel Core i7 processor, which is significantly more powerful than typical resource-constrained devices that are found in the DWC environment.

In another research Raveendra et al. [74] proposed a lightweight authentication protocol that facilitates mutual authentication and session key agreement between EV, CP and CSP. It leverages the concept of PUF to achieve ultra-lightweight authentication and provide resilience against protocol suite offers machine learning attacks. The security of the protocol has been validated using the Random Oracle model and Tamarin tool. However, the system is still vulnerable to impersonation, attacks by privileged insiders and key leakage. In addition, the protocol does not offer forward secrecy.

Bianchi et al. [75] presented a lightweight authentication protocol that achieves session security by creating pseudonyms through the utilization of ID-based public key infrastructure, Weil Diffie–Hellman (WDH) and the Boneh-Franklin system, which combines the benefits of public-key encryption and key generation. This scheme offers an on-demand charging request feature that allows users to halt the charging process based on their preference, eliminating the requirement to predefined the desired amount of power at the start of the process. This approach consists of a secure billing scheme as well. The protocol is resilient against miscellaneous attacks like free-riding and double-spending. Security analysis has been conducted using the Scyther tool and a MIRACL-based testbed experiment.

Huang et al. [76] proposed a lightweight authentication protocol between EVs and fog servers in DWC. Lightweight hash functions and XOR functions are the basic cryptographic primitives they have used to achieve lightweight and minimal computational demand. By using BAN logic, authors have demonstrated that the scheme successfully facilitates mutual authentication and is resilient against impersonation,

Table 6
Anonymous authentication protocols in dynamic EV charging.

Ref Year	Used primitives	Achieved properties	Entities	Phases	Limitations
[67] 2020	Pseudonyms Hashing Authenticated pairwise keys	+ Fast and lightweight authentication + Anonymous authentication + Preserves identity and location privacy	EV, TA CP Bank, RSU	1. Verifiable pseudonyms pre-distribution 2. Payment authorization 3. Anonymous authentication and recharging permission 4. EV Charging process 5. Billing and redemption	- Completely depend on a trusted authority for registration and pseudonym generation. - Continuous pseudonym requests from various EVs exhaust the resources of TA its and efficiency. - No clues of a security analysis conducted to verify the security of the protocol.
[68] 2020	Blind signatures Partial message proof Implicitly authenticated Diffie-Hellman key exchange Threshold secret sharing scheme	+ Revocable anonymity based authentication	EVs CS Financial authority (CA) Department of Motor Vehicles	1. Registration 2. Pseudonym collection 3. Token Collection 4. Permit Collection 5. Report and receipt generation 6. Anonymity revocation	- Lacks comprehensive information on how to achieve privacy preservation.
[69] 2021	Blockchain (smart contracts) Distributed-PKI Zero-knowledge proof Ring-signature superposition	+ Completely anonymous verification + Eliminate the need for trust of and dependence on EVSPs + Privacy protection + Undeniability + Unforgeability	EV Users PKI EVSP Charging station RBC CBC	1. Registration 2. Charging scheduling 2.1 Submit charge application 2.2. Deploy the smart contracts on the CBC 2.3 Ring-sign 3. Charging payment	- Lack of proofs on security of the scheme against attacks.
[70] 2023	Blockchain Group signature scheme	+ User anonymity + Data unlinkability + Security against DoS attacks + Include billing	EV, CP CSP CA Bank Blockchain network	1. Initialization 2. Charging request 3. Scheduling 4. Authentication and charging 5. Payment	- High computation and communication costs. - Less suitable for resource-constrained environments.

replay, and known session key attacks. However, the authors have not tested The system's ability to maintain efficiency in large-scale deployments.

Lee et al. [77] presented an enhanced authentication protocol utilizing PUFs which are resilient against various PUF-specific attacks. As Ascon has been selected by NIST standard lightweight schemes, to comply with resource-constrained environments and the lightweight, authors have employed the Ascon cipher suite in their protocol. To achieve the confidentiality and integrity of messages Ascon-128 authenticated encryption and Ascon-Hash as the cryptographic hash function are used. The authors have presented a detailed cryptanalysis of several PUF-based DWC authentication protocols and their resilience against those attacks. However, the assumption that the CSP is a trusted entity raises concerns in practical scenarios. Unconditional trust kept on CSP is practically undesirable, as it allows the service provider to track user behaviour which may violate the privacy of the EV. The summary of lightweight authentication protocols is depicted in Table 7.

4.1.4. Efficient authentication protocols

An efficient authentication protocol in a DWC system is a security framework tailored to meet the high-speed and real-time communication demands of EVs in motion during a dynamic charging session. The key focus of these protocols is to ensure seamless and continuous authentication as EVs move between RSUs and CPs, thereby enabling uninterrupted charging services. Efficiency in this context implies minimal delays in the authentication process, providing a rapid and reliable means for the EV to establish its identity and charging credentials. These protocols must be designed to swiftly and accurately validate the authorization, supporting the dynamic nature of the charging process. In essence, Efficient Authentication Protocols in DWC systems focus on optimizing real-time authentication speed and accuracy, ensuring that EVs can seamlessly transition between charging infrastructure components while maintaining efficient and secure charging services.

Li et al. [79] formulated a key dissemination-based fast authentication technique, which ensures message integrity. Acting as the central hub, RSU just forwards the messages from the EV to the utility after verifying the authenticity of the EV and vice versa. As the charging is performed while moving, the vehicle is expected to be authenticated with many RSUs situated along his driveway by creating unique session keys each and every time. This continuation of authentication accounts for a non-trivial computational cost and latency. To overcome the complication of re-authentication, it broadcasts the session key with the adjacent RSUs in the drive path. To avoid the congestion that might occur in the RSU, they used a queue to prioritize the key generation messages over application messages. However, in this scheme authors have not considered addressing the confidentiality of the messages and the privacy of user identity. In case of malicious compromise of the broadcasted session key takes place, the sensitive information of the user will be violated.

Rabieh at el. [80] have presented an efficient and privacy-preserving authentication scheme without the involvement of a trusted third party. The scheme consists of EVs, CSPs and a charging management system (CMS) which is responsible for controlling a set of charging controllers (CCs). Unlike the other authentication schemes in the same domain, this scheme does not require the participation of an RSU for the authentication between the EV and the charging pad. The protocol uses crypto primitives such as blind bilinear pairing-based signatures to anonymously authenticate the EV to the charging station and hash chains to perform the authentication between charging pads and the EV. Authors have accounted for the computational resource limitations when determining the appropriate cryptographic techniques. The MIRACL crypto library has been used to calculate the computational times

Table 7
Lightweight authentication protocols in dynamic EV charging.

Ref Year	Used primitives	Achieved properties	Entities	Phases	Limitations
[78], 2019	Symmetric cryptography XOR Hash	+ Location privacy Anonymity + Include a payment scheme + Uses 5G cellular networks	EV CP CSP Fog server RSU	1. Pre-registration 2. Registration 3. Authentication 4. Charging 5. Payment	- TA involved in each charging session - Vulnerability to free riders. - Unfair billing due to the inefficiency of charging - Lack of ability to trace the payments.
[71] 2021	Elliptic curves Hash chains XOR	+ Minimal computational and communication cost + Confidentiality	EV, CP Charging server-CSP Fog server RSU	1. FS registration 2. EV registration 3. Login 4. Authentication	- Suffering from fog-server impersonation and replay attacks. - Decentralized setup and multiple rounds of mutual authentication increase cost.
[73] 2022	PUF LFSR stream cipher Espresso Blockchain Smart contract Pseudonyms	+ Identity anonymity + Data integrity + Information hiding + Lightweight	EV CSP RSU CP Blockchain	1. Initialization 2. Registration 3. User authentication 4. Login + mutual authentication 5. Handover	- Lack of key revocation. - Ignored the price optimality aspect. - Higher comm. and computation cost.
[75] 2022	Bilinear pairing Hash functions ID-based public key encryption	+ Computationally lightweight + Fast authentication	EV, RA Bank CSP, RSU CP Fog server	1. System initialization 2. EV registration 3. Authentication of CSP, RSU, CP 4. Termination	- No consideration of user privacy and anonymity.
[72] 2022	Hash XOR Concatenation MAC	+ Lightweight + Fast authentication + Seamless Handover + EV Revocation + Functions to dynamically add RSUs and EVs	EV CSP Fog server RSU CP	1. Initialization 2. Registration 4. Login 5. Mutual authentication 6. Handover 7. EV Revocation	- Lack of revocation techniques (only self revocation). - At least 1 to 2 system entities know the original identity of EV. - Multiple mutual authentication rounds are required. - Vulnerable to DoS attacks.
[74] 2022	PUF XOR Hash	+ Physically safe and resilient against different types of attacks including ML attacks.	EV CP RSU CSP RCS	1. Initialization 2. Registration(RCS,EV) 3. User authentication 4. Login, authentication 5. Handover	- Vulnerable to impersonation, privileged insider and key leakage attacks. - Lack of forward secrecy. - CSP is considered as a trusted entity.
[76] 2023	Lightweight hash functions XOR	+ Anonymity +Untraceability + Resilient against different attacks	EV, CP Charging server-CSP Fog server RSU	1. FS registration 2. EV registration 3. Login 4. Authentication	- The system's ability to maintain efficiency in large-scale deployments is not explicitly addressed.
[77] 2023	PUF Ascon-128 Ascon-Hash XOR	+ Resilient against different types of attacks + Forward secrecy + Confidentiality + Integrity	EV TSP RSU RCS CP	1. One-time enrolment 2. Device authentication and key exchange * Node to Node (N2N) * Node to Server (N2S)	- TSP/CSP is assumed to be trustworthy.

of the protocol. Even though the scheme is resistant to double spending and collusion attacks no proper proof is provided against the major attacks such as impersonation, injection and denial of service attacks.

Nguyen et al. [81] formulated authentication and physical layer security schemes enabling secure communications between the EV and CP. A double-encryption with the signature (DoES) is used for session key exchange which provides data authenticity and integrity. A symmetric key-based sign-encrypt-message (SEM) authentication scheme is used to achieve low-latency authentication between EV and power transmitter (PT)/CP. For security at the physical layer, an artificial noise-based physical layer security (AN-based PLS) scheme is included in the protocol. Although the SEM scheme is specially designed to achieve low-latency authentication, the combination of SEM and the DoES still introduces latency. In high-demand environments where multiple EVs are simultaneously trying to authenticate and start the charging session, even small latencies could accumulate, affecting system efficiency. A summary of the efficient authentication protocols in dynamic EV charging is illustrated in Table 8.

4.2. Billing and payment protocols

Secure billing/payment protocol is a critical component of a dynamic EV charging system that ensures reliable and trustworthy financial transactions between CSPs and EV owners. The dynamic nature of EV charging, where vehicles connect and disconnect from charging utilities frequently involves a charging "handover" from one charging utility to another, introduces unique challenges in maintaining billing continuity and payment security. These challenges significantly impact the operations and management of the wireless charging infrastructure. To address these challenges, researchers study many paths to achieve goals like privacy preservation, anonymity and lightweight. These approaches involve the use of secure authentication and authorization mechanisms by employing cryptographic techniques such as zero-knowledge proofs and blockchain techniques to verify the authenticity of charging stations and EV owners. By ensuring that only authorized entities can participate in the payment process, the risk of fraudulent activities and unauthorized access to payment information can be mitigated. Some approaches highlight the use of tamper-resistant charging hardware and metering systems. These systems employ secure hardware components and cryptographic algorithms to

Table 8
Efficient authentication protocols in dynamic EV charging.

Ref Year	Used Primitives	Achieved Properties	Entities	Phases	Limitations
[79] 2015	HMAC ECDSA Just Fast Keying (JFK)	+ Efficiency + Security	Certificate authority Charging Utility RSU	1. Session key establishment with RSU 2. Broadcasting session key with RSUs 3. Establish priority queue to address authentication	- Message confidentiality not achieved - Broadcasting the session key between RSUs - Vulnerable to replay and eavesdropping
[80] 2017	Blind signature Signed hash chains	+ Efficient, privacy preserving without a TTP + Computationally cheap + Fit-in with limited resources device	EVs Charging pads Charging station CSP Charging management center Charging controllers	1. System bootstrap and tickets purchasing 2. Anonymous charge permission 3. Charging request and reply 4. EV charging	- Proper proofs for security analysis are not provided against the major attacks such as impersonation, injection and DoS attacks.
[82] 2022	Hash functions XOR Hash chains El-Gamal over elliptic curve	+ Efficiency + Security	EV CSP Authority Vehicle registration and revocation authority (VRRA) Charging Pad	1. Registration 2. Authentication phase 3. Charging phase	- Centralized authentication infrastructure. - High computational overhead at CSPA.
[83] 2023	XOR Hash chain Pseudonyms	+ Low complexity + Secure authentication + Secure against replay attacks	EV CP CSPA RA	1. OBU registration 2. Pseudonyms generation 3. Initialization 4. EV-CSPA authentication	- Considered only the replay attack and DoS attack.
[81] 2023	Double encryption signature (DoES) Sign-encrypt-message (SME) AN-based PLS	+ Secure session key exchange + Low-latency + Authentication and physical layer security scheme	EV CSP/CS CP RSU PT	1. Session key exchange with DoES 2. Low-latency authentication with SEM 3. Billing	- High computation cost due to DoES. - Although the schemes are designed to achieve low latency, the combination of SEM and DoES still introduce latency.

protect against physical tampering or manipulation of charging stations and meters. This helps ensure accurate measurement of energy consumption and prevents unauthorized modification that could lead to fraudulent billing practices. Fair billing and payments of wireless EV charging lanes are key complications of implementing the facility as a commercial service. Researchers follow various billing models to satisfy the requirements above mentioned. An example of such a strategy is to base the billing on the actual amount of energy received by each EV, rather than the duration of time the EV spends on the wireless charging lanes [84].

Zhao et al. [85] proposed a secure and privacy-preserving billing scheme for online EVs (OLEV) systems. Scheme uses a trusted platform module (TPM) embedded in the OBU which is assumed as a completely trusted and tamper-resistant entity to hold the identity of the EV. This billing scheme operates under the assumption that each charging pad delivers a consistent quantity of energy to the EV and the CSP charges the EV the same amount for the electrical energy received from every charging pad. For the payments, the user should top his accounts before the charging session. Authors have used GUN multi-precision and pairing-based crypto library-based testbeds to calculate the computational time and state that the scheme is secure against replay attacks. However, the proposed scheme has a high computational cost as it has used a TPM, which is not satisfactory for a fast-paced environment like dynamic EV charging.

Mojtaba et al. [78] proposed a payment scheme, that initiates with a preregistration phase that employs symmetric keys, followed by hashing together with the various charging segments. Billing is performed on a segment-by-segment basis, which involves a bidirectional audit conducted between the CSP and the EV, with the assumption that a fixed quantity of energy is provided per segment. Once the charging process has been completed, the EV communicates the amount of energy it has received to all relevant entities. Subsequently, the EV is billed based on the transferred energy level. The scheme is considered resistant to replay, MITM, inside and impersonation attacks. Formal security analysis has been conducted using the AVISPA tool and the informal analysis as a collation of existing protocols. A notable drawback of segment-by-segment billing used in this scheme is, the susceptibility to free-riders exploiting the handover delay between segments, allowing them to charge their vehicles both before and after authenticated EVs. In each charging instance, the TA becomes involved in the communication between the EV and the CSP resulting in an additional communication overhead. The protocol cannot trace charging session payments, which compromises the ability to verify the fairness of the billing process.

Ahao at el. [86] proposed a privacy-preserving billing scheme for DWC systems that is resilient against free riders. In this scheme, authors have addressed this issue of segment-by-segment billing schemes by proposing a privacy-preserving billing approach. The proposed approach employs token authentication and tamper-proof meters to enhance privacy and security. Each authenticated EV is required to securely share its power levels before and after passing through the charging lane and these readings are then compared to the actual energy supplied by the charging pads. By analysing the disparity between the recorded power levels and the supplied energy, the system can effectively identify instances of free-riding while simultaneously preventing the over-billing of authorized EVs. This ensures that only legitimate charging sessions are accounted for, addressing the concern of unauthorized usage and ensuring fair and accurate billing for authenticated EVs. EVs are expected not to occupy the same charging segment, which is defined as being 50 m in length or potentially extending over

Table 9
Payment and billing protocols in dynamic EV charging.

Scheme Year	Used Primitives	Achieved Properties	Entities	Phases	Limitations
[85], 2016	Zero-Knowledge Proof Range Proof Bilinear Maps Camenisch-Lysyanskaya (C-L) Signature	+ Privacy-preserving + Can charge EVs anonymously + Can offers discounts to special users	EV Charging Service Company (CSC) Initialization Authority (IA) Power transmitter (PT)	1. Initialization 2. EV registration 3. Account recharge 4. Power purchase 5. Battery recharge	- Has a high computational as using a TPM.
[86], 2017	Digital signatures Hash Function	+ Computationally lightweight + Fast authentication	EV RA Bank CSP	1. Token obtaining 2. Token using 3. Redeeming	- No consideration of user privacy and anonymity
[78], 2019	Symmetric Cryptography XOR Hash	+ Lightweight, security + Location privacy + Anonymity + Uses 5G cellular network for exchange of heavy traffic + Avoid free riders	EV CP CSP Fog server RSU	1. Pre-registration 2. Registration 3. Authentication 4. Charging 5. Payment	- TA involves in each charging, resulting an additional overhead). - Vulnerability to freeriders - Potential for unfair billing due to the inefficiency of charging segments employed.
[88] 2023	Blockchain Smart contracts Zero-knowledge proofs	+ Privacy and unlink-ability of transactions	EV, CP CSP Smart contract	1. Initialization 2. Charging request 3. Credentials issuing 4. Authentication 5. Charging and payment	- High computation cost due to heavy crypto techniques. - Non-trivial communication cost.

a mile, to avoid the free riders. According to this scheme, all vehicles within a traffic jam would be obligated to pay for charging, irrespective of whether they require a charge or not. Thus, the proposed approach becomes impractical.

Hussain et al. [87] proposed fairness in billing and authentication for OLEV utilizing wireless charging. This proposed framework operates on the assumption that the power track comprises segments, and each segment within the track transfers a consistent amount of energy to the EV wirelessly. Billing is conducted on a segment-by-segment basis. Authors assume that each EV is equipped with a tamper-resistant module responsible for carrying out secure calculations necessary for authentication and billing. This module facilitates a bidirectional audit between the CSP and the EV. The main drawback of this scheme is that the CSP can associate the authentication data and deduce the identities of the EV owners.

Massmi et al. [88] proposed a secure and fair billing scheme for the DWC system, based on smart contracts. The payment scheme includes a lightweight authentication mechanism that enables continuous authentication for the charging process when the EV is moving on the power track. By deploying the smart contracts within a blockchain network, this approach ensures a decentralized and verified billing process that does not depend on centralized intermediaries, consequently enhancing privacy. However, the complex design with sophisticated cryptographic primitives accommodates a significant computational demand and substantial communication overhead. A summary of the payment and billing protocols in dynamic EV charging is illustrated in Table 9.

4.3. Coordination and management

In addition to security considerations, authentication is crucial for effective energy management, coordination, lane management, scheduling and navigation management in dynamic EV charging systems. Authentication allows CSPs to track (if necessary) and manage the charging activities of individual EVs, enabling efficient scheduling of charging sessions. By accurately identifying and authenticating EVs, the charging system can assign appropriate charging slots, optimize the allocation of resources, and ensure a balanced distribution of charging demand. Authentication also facilitates advanced energy management techniques, such as load balancing and demand response, by providing real-time information on the charging status and energy requirements of individual EVs. This enables CSPs to dynamically manage and allocate energy resources, optimize the utilization of available infrastructure, and prevent overloading or congestion in the charging network. Furthermore, authentication can enable personalized billing and tariff structures, tailored to the specific requirements and preferences of individual EV owners, promoting fairness and transparency in the charging process.

Effective functioning of the charging infrastructure necessitates the integration of an ICT system, responsible for managing lane users, processing booking requests, and issuing billing statements in dynamic charging environments [89]. To facilitate the navigation of EVs towards the appropriate charging lanes, proper navigation can be utilized. This navigation system may inform users about the precise location of the charging lanes, the corresponding charging costs, and lane availability for serving their EVs. Moreover, the navigation system employs advanced algorithms to optimize energy consumption and devise efficient charging plans for EV journeys.

Xiaofeng et al. [90] formulated an energy harvesting DWC (EH-DWC) protocol. The proposed system involves EV registration and power negotiation with the charging controller, along with hierarchical token-based authentication facilitated by RSUs. This system operates under the assumption that the charging energy is obtained from renewable energy sources (RES), making power negotiation crucial to ensure the availability of energy. To implement this, the system utilizes pseudonyms and digital signatures for the initial registration of EVs with the PSS. Taking into account that varying weather conditions may affect energy harvesting and system operation management, the scheme introduces three distinct states: high power state, medium power state, and low power state. Each state is designed to facilitate efficient authentication, secure communication, reliable payment, and protection of the identity between the system and the EV. This approach ensures adaptability to different conditions while maintaining the system's functionality and privacy measures. Even though a security analysis has been conducted, it is only tested upon replay and MITM attacks. No indication has been provided about the resistance of this scheme against impersonation and denial of service attacks which is un-negligible in DWC environments.

Table 10
Security measures for DWC authentication.

Security measure	Acronym	Description
Anonymity	S1	Ensure an attacker cannot extract the ID of the EV user by intercepting messages transmitted in an insecure communication channel, and the attacker may not even be capable of finding any relation between two specific sessions [91]. This preserves the privacy of the EV by preventing unauthorized identification or profiling of the EV based on charging habits.
Non-traceability	S2	Verify that During the authentication, any information should not be exchanged that makes the user traceable to either a CP, RSU or an eMSP, preventing the creation of a user's movement profile [92]. This assures the EV's privacy and prevents the EV from being traced by unauthorized entities.
Mutual authentication	S3	Ensure that, both the EV and the charging infrastructure should authenticate each other before start exchanging messages. It makes sure that the EV is connected only with a trusted charging infrastructure for charging.
Session key agreement	S4	Ensure confidentiality of the exchanged messages by establishing a shared secret key between the EV and the charging station for a specific charging session [93]. It ensures that the sensitive data exchanged (vehicle ID, payment information, charging status, EV location) and the communication sequence remains confidential.
Perfect forward secrecy	S5	Ensure that the past charging session keys remain secure even if the long-term secret keys of the EV or other entity are compromised [94]. Thus the encrypted past communication cannot be able to be decrypted and any leakage of a long-term key should not reveal the historical data of the EV.
Location privacy	S6	Verify that an EV's current location or the charging route is not disclosed during the charging session [95]. This prevents attackers from tracking the real-time location of the EV and further security risks which can arise due to the location tracking.

4.4. Discussion of cryptographic techniques employed in authentication protocols for DWC systems

In this section, we explore some of the essential security measures in DWC authentication to preserve privacy and achieve security such as anonymity, non-traceability, mutual authentication, session key agreement, perfect forward secrecy and location privacy. Authentication protocols in DWC should incorporate these measures to ensure a high level of security and privacy for all participants. Table 11 illustrates a comprehensive study of different approaches taken to design state-of-the-art authentication protocols, and how they address essential security measures (see Table 10).

According to Table 11, blind signatures are a widely used cryptographic technique to achieve full anonymity. It allows the user to obtain a valid signature from a signer without revealing the content of the message to the signer. In dynamic EV authentication, blind signatures are used to authenticate the EV without revealing the true identity to the untrusted parties like CSP, RSU or CPs. PUFs are used in dynamic EV charging authentication to securely identify and authenticate the EVs to the other entities who take part in the communication [96,97]. PUFs are electronic circuits that are designed to produce unique responses to specific challenges or inputs, which are difficult to replicate or clone. In the context of dynamic EV charging authentication, PUFs can be integrated into the charging pads as well as the EV itself and the EV is required to generate and exchange unique authentication keys that can be used to securely authenticate the devices and authorize the charging process. However, this technique is completely depending on the hardware component and it requires a completely new hardware infrastructure to make use of this technique. Zero-knowledge proofs are a cryptographic technique that allows a party to prove the authenticity of a claim without revealing any information about the claim itself [98]. In the dynamic EV authentication, authors of [69,99] have demonstrated using zero-knowledge proofs to achieve anonymity.

Despite the use of anonymous authentication to ensure identity and location privacy, it is considered that anonymous authentication techniques may lead to malicious insider attacks [100]. Any EV, which got authenticated as a legitimate user, can perform different types of attacks. For instance, a malicious EV can exploit this vulnerability, by continuously sending a massive amount of legitimate charging requests to the RSU/charging utility, while refraining itself from charging. This violates the availability of charging stations by denying the service of the infrastructure to other legitimate EVs travelling in the same area. In this context, as the identity of the misbehaving EV is completely hidden from the charging infrastructures, the system does not have any mechanism to distinguish the malicious EV and to take necessary steps to prevent further attacks from the particular EV. Any malicious EV can unambiguously attack a charging station and remain undetected, pretending as a legitimate user.

5. Research challenges and open problems

Upon an extensive analysis of a wide range of authentication protocols in the domain, we have identified some research gaps which are not yet received significant attention, but hold a great value. As some directions for future researchers, this section explains those identified gaps that may lead to new directions of research and ultimately enhance the betterment of the domain.

5.1. Utilizing blockchain technologies

For the dynamic EV authentication, researchers use blockchain techniques to store digital certificates and other authentication-related data in a decentralized and tamper-proof manner. Thereby researchers effort to acquire the characteristics of the blockchain such as decentralization, security, immutability and transparency [65,101]. Many researchers for instance, [100,102] have used blockchain to achieve privacy as well as anonymity in the context. A blockchain is a decentralized and distributed ledger that maintains a transparent and static record of transactions. In the dynamic EV authentication context, the blockchain is used to store and verify authentication-related sensitive information, such as user vehicle identity, digital certificates, location details, etc. As these records are stored in blockchain, it verifies that the information cannot be tampered with. Also, each charging session can be recorded as a transaction on the blockchain, ensuring the integrity and immutability [103] of sensitive data which prevents unauthorized access or manipulation of charging data, ensuring the accuracy of billing and preventing malfunctions.

As the blockchain is decentralized in nature, multiple nodes in the network are required to validate the integrity of the transactions using different consensus algorithms such as proof of work (PoW) and proof of stake (PoS). Thus, it eliminates the single point of failure, making it more attack resilient and verifies that a single entity cannot

Table 11
Achieved security measures and the underlying cryptographic primitives of authentication protocols.

Ref, Year	Type of authentication	Crypto primitives	S1	S2	S3	S4	S5	S6	Informal proofs	Formal proofs	Trusted entities
[59], 2015	Hash chain-based mutual authentication	Hash chain, Pseudonyms, XOR	✓	✗	✓	✓	✓	✗	✓	–	2 or more
[60], 2017	Pre-shared key based authentication	PKI, Pseudonyms, MAC, Symmetric key cryptography	✗	✗	✓	✓	✓	✗	✓	–	2 or more
[61], 2018	Token-based authentication	XOR, Hash functions, Partial blind signatures	✓		✓	✗	✗	✓	✓	–	1
[62], 2020	Token-based authentication	ECDH, Short and blind signatures, Bilinear pairing, Hash chain	✓	✗	✓	✓	✓	✓	✓	AVISPA	1
[63], 2021	Chaotic map based authentication	Chaotic map (Chebyshev), Blind signature, Hash function, PRNG	✗	✗	✓	✓	✓	✗	✓	–	1
[64], 2021	Blockchain-based authentication	Blockchain, PRNG HMAC, Group signature	✓	✗	✗	✗	✓	✗	✓	–	2 or more
[66], 2023	Token-based authentication	ECDH, Blind Signatures, Bilinear pairing, Hash chain	✓	✗	✓	✓	✓	✗	✓	–	2 or more
[67], 2020	Pseudonym-based authentication	Pseudonyms, Hash function, Camenisch's proof of knowledge	✓	✗	✗	✓	✗	✓	✗	–	2 or more
[68], 2020	Certificate-based authentication	Blind signatures, Zero-knowledge proof, Implicitly authenticated DHKE,Threshold secret sharing	✓	✓	✗	✓	✓	✗	✓	–	2 or more
[69], 2021	Certificate-based authentication	Blockchain, Distributed-PKI, Zero-knowledge proof, Ring-signatures	✗	✓	✗	✗	✗	✗	✓	–	2 or more
[70], 2023	Blockchain-based authentication	Blockchain, Group signature, Zero-knowledge proof, Hash function,	✗	✓	✗	✗	✓	✗	✓	–	–
[78], 2019	Pre-shared key based authentication	Kerberos version 5, XOR, Hash function	✓	✓	✓	✗	✗	✗	✓	AVISPA OFMC	3 or more
[71], 2021	Pre-shared key based authentication	ECDSA, Hash chains, XOR	✓	✓	✓	✓	✓	✗	✓	Scyther ROR	2 or more
[73], 2022	PUF-based authentication	PUF, Stream cipher Espresso, Blockchain, Pseudonyms	✓	✗	✗	✗	✗	✗	✓	AVISPA	2 or more
[75], 2022	Identity-based authentication	Bilinear pairing, Hash functions, ID-based PKE	✓	✗	✓	✓	✓	✗	✓	BAN logic	2 or more
[72], 2022	Pre-shared key based authentication	Hash, XOR, MAC	✓	✓	✓	✗	✓	✗	✓	Scyther ROR	2 or more
[74], 2022	PUF based authentication	PUF, XOR, Hash function	✓	✗	✓	✓	✗	✗	✗	Tamarin ROM	1
[76], 2023	PKI based authentication	Hash functions, XOR	✗	✗	✓	✓	✓	✗	✓	BAN logic	2 or more
[77], 2023	PUF based authentication	PUF, Ascon-128, Ascon-Hash, XOR	✓	✓	✓	✓	✓	✗	✓	–	2 or more
[79], 2015	Just-Fast-Keying based authentication	PKI, HMAC, ECDSA, JFK	✗	✗	✗	✓	✓	✗	✗	–	2 or more
[80], 2017	Token-based authentication	Blind signature, Signed hash chains	✓	✓	✓	✓	✓	✓	✓	–	0
[82], 2022	Pseudonym-based authentication	Hash functions, XOR, Hash chains	✓	✓	✓	✗	✗	✓	✓	BAN logic Scyther	2 or more
[81], 2023	Signcrypion based authentication	Double encryption signature, Signature-encrypt-message (SME)	✓	✗	✓	✓	✓	✗	✓	–	2 or more
[85], 2016	ZKP-based authentication	Zero-knowledge proof, Range proof, Bilinear maps, Camenisch-Lysyanskaya Signature	✓	✓	✗	✗	✗	✓	✓	–	2 or more
[88], 2023	Blockchain-based authentication	Blockchain, Zero-knowledge proof	✓	✓	✗	✗	✗	✗	✓	AVISPA	–

Legend for the table: S1: Anonymity S2: Non-traceability, S3: Mutual Authentication, S4: Session Key Agreement, S5: Perfect Forward Secrecy, S6: Location Privacy.

manipulate the authentication process [104]. Also, it eliminates the need for the presence of a central authority or trusted third party for the authentication process, as practised in many centralized authentication schemes. Blockchain enables peer-to-peer transactions between EV owners and charging station operators without the need for intermediaries (utility companies) [105]. EV owners can securely interact with charging stations, authenticate themselves, and authorize charging sessions directly through the blockchain. This decentralized approach reduces reliance on centralized systems and increases efficiency by eliminating unnecessary intermediates as it does not require trusted third parties for the authentication.

Blockchain technology also provides strong security mechanisms [106–108]. Along with blockchain, advanced cryptographic techniques are used to secure the information stored on the blockchain. These

mechanisms help protect user identities, certificates, and authentication data, reducing the risk of unauthorized access or identity theft. By utilizing blockchain technology in dynamic EV authentication, the security and reliability of the authentication process can be enhanced, protect user identities, and establish a trustworthy and auditable system that provides an extra layer of assurance and mitigates the risks associated with traditional centralized authentication systems. However, due to the highly complex nature of the blockchain architecture, it possesses certain limitations. For instance, consensus protocols demand high computational power and they do not provide efficient processing of the transactions [109]. Thus, utilizing the advanced security features in blockchain to perform efficient authentication with low computational and communication costs is valuable to be further explored.

5.2. Using homomorphic encryption to achieve location privacy

Homomorphic encryption (HE) is a widely used tool for incorporating privacy preservation into distributed control systems [110]. It allows for secure arithmetic operations to be performed on encrypted data, and the output obtained after decryption corresponds to the computations conducted on the initial plaintext, [111–114]. A prevailing trend in inductive EV charging scenarios is to use Homomorphic encryption to protect the privacy of user data while enabling certain computations to be performed on the encrypted data [115–118]. Location privacy is a crucial thing in any system, which may lead to many severe attacks in case of a dilution [119–121]. During the communication between the entities in the charging environments, a variety of personal information such as unique user/EV identity, state of charge, location information, and payment information is exchanged between the EV, RSU and the charging utilities [30]. This information can be utilized to derive the location and travel behaviours of the EV. Communication entities involved in dynamic EV charging, are the potential for a massive number of threats, including impersonation, replay, man-in-the-middle attacks, guessing, and hijacking [37]. As a solution, privacy-preserving authentication protocols are used to authenticate itself to the CSP without revealing its true identity. Homomorphic encryption has become an intelligent choice for them. The aforementioned research works to explain how homomorphic encryption has been used in conducting EV charging scenarios to achieve location privacy. Applying homomorphic encryption in dynamic inductive charging authentication would be a revolutionary concept as location privacy is an ultimate goal in this context.

5.3. Seamless handover and continuous authentication

Seamless handover is not a new concept to the industry, as it is being broadly used in a variety of applications such as mobility devices, mobile vehicular networks, etc [122–124]. Seamless handover and continuous authentication in DWC mainly focus on a secure and uninterrupted charging experience as EVs move between different RSUs and charging pads during the ride. These functionalities are crucial for ensuring efficient and reliable charging services in a DWC environment. Seamless handover involves the smooth transition of an ongoing charging session from one RSU/charging pad to the other, allowing EVs to maintain a continuous power supply without any disruptions or the need to restart the charging process. This requires coordination and communication between charging stations to exchange relevant information about the charging session, such as charging parameters, payment details, and authentication credentials. Continuous authentication is an essential aspect of maintaining the security and trustworthiness of the charging process. It involves verifying the identity and authorization of the EV and the charging station throughout the charging session, even during handovers [125]. This authentication process helps prevent unauthorized access, ensures that the charging session is only initiated and continued by authenticated entities, and protects against potential security threats or malicious activities. However, upon examining the existing research literature, it becomes evident that there is a limited amount of research work specifically dedicated to this particular context.

5.4. Anonymous authentication and revocation of malicious EVs

Another popular subject in dynamic EV charging is anonymous authentication. Distinct cryptographic techniques are used by recent researchers to achieve complete anonymity in their authentication protocols. Blind signatures, physical unclonable functions, Zero-knowledge proofs, and various blockchain techniques are mainstream. Despite the use of anonymous authentication to ensure identity and location privacy, it is considered that anonymous authentication techniques may lead to malicious insider attacks. [100]. An EV that has been verified as an authorized user has the potential to execute various forms of attacks. For instance, a malicious EV can exploit this vulnerability, by continuously sending massive amounts of legitimate charging requests to the RSU/charging utility, while refraining from charging. This violates the availability of charging stations by denying the service of the infrastructure to other legitimate EVs travelling in the same area. In this context, as the identity of the misbehaving EV is completely hidden from the charging infrastructures, the system does not have any mechanism to distinguish the malicious EV and to take necessary steps to prevent further attacks from the particular EV. Any malicious EV can unambiguously attack a charging station and remain undetected, pretending as a legitimate user. To resolve the vulnerabilities arising due to anonymous authentication, robust revocation mechanisms should be embedded into the authentication phase. Revocable anonymity-enabled authentication allows the system to distinguish or trace back the users, in case of malicious behaviour [68]. In the dynamic EV charging context, currently, there is a lack of such protocols. Even the existing revocable anonymity-based authentication protocols have various security flaws like the potential for application-level denial of service attacks [126]. This background signifies the necessity of having functionalities to reveal the true identity of the user upon a distinguished malicious activity within the system.

5.5. Challenges of dynamic EV charging environments

Not only privacy preservation and anonymity there are other security-related goals that a successful authentication protocol should consist of in these settings. Especially, the authentication protocol should adequately address the nature of the context. A dynamic charging environment is a fast-phase environment in which the EV is charged while moving. Thus, any cryptographic technique applied, ought to perform very first to execute the authentication while the EV is moving firstly in-between entities such as RSUs and CPs. While the advanced cryptographic algorithms have a non-trivial execution time with multiple authentication rounds, efficient and lightweight schemes may lead to potential security issues. So, it is essential to find the balance between the used cryptographic primitives, achieved security level and efficiency. Furthermore, the resource-constrained nature of this environment, coupled with the limitations of the OBUs and other devices within the system, presents an additional layer of complexity. Implementing secure authentication algorithms under such resource-constrained conditions poses a substantial challenge. The need to preserve not only privacy and anonymity but also the availability and integrity of the system's services adds to the intricacy. As a result, the design and implementation of authentication protocols in dynamic EV charging environments demand a judicious consideration of the inherent resource constraints, reinforcing the need for efficient, lightweight, and secure solutions tailored to this unique setting.

5.6. Adhering with existing standard protocols like ISO 15118

Different charging strategies and supporting communication protocols are currently in use with static charging stations such as ISO15118 and Chademo. ISO15118 is the international communication standard for static plug and charging which enables integrating EVs into the smart grid. Edition 1 of the protocol suit is the most widely used

to date and edition 2 is expected to increase adoption as it includes many functional enhancements. Edition 2: ISO15118:20 enables bi-directional communication between the EV Communication Controller (EVCC) and Supply Equipment Communication Controller (SECC) during the charging procedure and smart energy management. However, the ISO 15118 protocol currently supports only static, plug-and-charge operations. As the ISO15118 standard is the most widely used and well-established protocol, incorporating it into new authentication protocol designs is more practical for real-world applications. Instead of creating entirely new authentication protocols, designing them to align with existing protocols in the current EV charging domain while utilizing new cryptographic combinations ensures practicality and maintains agility.

5.7. Use of advanced authentication techniques to enhance security

Most of the state-of-the-art authentication protocols are based on preliminary authentication techniques such as certificate-based, symmetric key, token or pseudonym-based approaches. Utilization of advanced authentication techniques such as attribute-based authentication or the design of hybrid authentication protocols combining multiple techniques can provide a balanced approach to achieve stronger security. However, the cost of such advanced cryptographic primitives may reduce the performance of the application due to the resource limitations in the EV, RSU and the CPs. Thus, the selection of appropriate primitives and the balance of the performance should be carefully addressed.

6. Conclusion

Dynamic EV charging is a novel technology which combines complex infrastructures, networks, communication protocols and different actors. As a result, the protocols are still in developing status as they lack crucial security measures such as user identity protection and location privacy. A comprehensive survey covering the technology, authentication protocols, and security requirements is unavailable in the current literature. Furthermore, the existing literature lacks a comprehensive summary of security metrics in authentication protocols and the cryptographic techniques employed. As a result of this lack, many existing protocols suffer from privacy and security issues. In this survey, we present the fundamentals of this new technology and dive into a new classification of state-of-art authentication protocols, an in-depth analysis of their strengths, weaknesses and the underlying cryptographic primitives to fill this gap. Further, we discuss a clear threat model and security requirements, existing vulnerabilities and potential countermeasures. Understanding the applicability of the crypto primitives to achieving the desired security measures in DWC systems is crucial for designing new protocols. As the cryptographic primitives are the fundamental building blocks of authentication protocols we ensure to thoroughly cover these aspects. We explored the different approaches taken to design the authentication protocols, yielded security measurements and utilized crypto primitives. Through our study, we revealed that many of the existing protocols have significant security and privacy issues. We also highlight the open challenges and the need for privacy-preserving and robust authentication protocols to safeguard against potential vulnerabilities. We also present the future directions and areas needing further work to assist researchers and industry professionals in designing new authentication protocols for DWC to address these gaps. To the best of our knowledge, this is the first research work to cover all those aspects of DWC systems. Our main findings are as follows.

Main Findings and Lessons Learnt

- Most of the existing authentication protocols are limited to the use of fundamental authentication techniques such as certificate-based, symmetric key, token or pseudonym-based approaches.
- Despite using a PUF, token or pseudonym-based approach to conceal the true identity of the user, these schemes still lack adequate location privacy. Many so-called anonymous authentication schemes use the same hidden identity throughout the journey (same pseudo identity with all the RSUs or CPs), which reveals the route of the EV. Even if the identity is not revealed, the attacker can still observe the driving behaviour of the EV.
- Privacy-preserving or maintaining anonymity during the authentication procedure should be rethought and more advanced crypto techniques should be utilized to conceal the true identity of the user and to achieve location privacy especially to hide the route of the EV.
- There is a lack of robust anonymous/privacy-preserving authentication protocols with seamless handover and advanced revocation capabilities.
- Use of heavy cryptographic operation during the authentication reduces the performance of the protocol as the DWC entities like EV, RSU and CPs are resource resource-constrained devices. Lightweight operations are much more suitable for such environments. However, the underlying cryptographic primitives should be carefully chosen without compromising security.

As the field continues to evolve, there is a critical need to address these challenges and develop innovative solutions that can enhance the security, efficiency, and performance of dynamic charging systems. These enhancements not only mitigate the identified limitations but also assure a resilient and user-friendly ecosystem for dynamic wireless EV charging. We think this survey paper serves as a valuable resource for researchers, protocol designers and security professionals involved in the development and deployment of authentication protocols for dynamic EV charging to strengthen the field further.

CRediT authorship contribution statement

Nethmi Hettiarachchi: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Saqib Hakak:** Supervision, Validation, Writing – review & editing. **Kalikinkar Mandal:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nethmi Hettiarachchi reports a relationship with New Brunswick Power that includes: funding grants. Kalikinkar Mandal reports a relationship with New Brunswick Power that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See Table 12.

Data availability

No data was used for the research described in the article.

Table 12
List of abbreviations.

Acronym	Description	Acronym	Description
BEV	Battery Electric Vehicle	LFSR	Linear Feedback Shift Registers
CA	Certificate Authority	MAC	Message Authentication Code
CMS	Charging Management System	MITM	Man-In-The-Middle
CP	Charging Pad	OBU	On-Board Unit
CSP	Charging Service Provider	PHEV	Plug-in Hybrid Electric Vehicle
CWD-WPT	Cloud Based Wireless Power Transfer	PKI	Public Key Infrastructure
DoS	Denial of Service	PoS	Proof of Stake
DWC	Dynamic Wireless Charging	PoW	Proof of Work
ECC	Elliptic Curve Cryptography	PT	Power Transmitter
ECDSA	Elliptic Curve Digital Signature Algorithm	PUF	Physically Unclonable Functions
ECDH	Elliptic-curve Diffie–Hellman	QDWC	Quasi-Dynamic Wireless Charging
EH-DWC	Energy Harvesting Dynamic Wireless Charging	RA	Registration Authority
EV	Electric Vehicle	RCS	Region Charging Server
EVCI	Electrical Vehicle Charging Infrastructures	RSU	Road Side Unit
EVCS	Electric Vehicle Charging Stations	SWC	Stationary Wireless Charging
FCEV	Fuel Cell Electric Vehicle	TTP	Trusted Third Party
FHE	Fully Homomorphic Encryption	V2G	Vehicle-to-Grid
GHG	Green-House Gas	VANET	Vehicular Ad-hoc Networks
HE	Homomorphic Encryption	WDH	Weil Diffie–Hellman
HEV	Hybrid Electric Vehicle	AVISPA	Automated Validation of Internet Security Protocols and Applications
HMAC	Hash-Based Message Authentication Code	ROR	Real-or-Random model
IoT	Internet of Things	BAN logic	Burrows-Abadi-Needham logic
JFK	Just Fast Keying	OFMC	on-the-fly model-checker
PRNG	Pseudo Random Number Generator		

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